# DCCREG turbine

## Design ledger and fact sheet

**The design lock of 2026-10-09, revised the same day after the settlement of its inconsistencies and open points.** A temporary documentation ledger and the current fact sheet of the exercise.

| | |
|---|---|
| status | **LOCKED for now** — the designer, 2026-10-09: "Let's lock the design for now." |
| design state | the lock's baseline is commit `09243c7` on branch `claude/new-session-0az7f9`. The settlement (2026-10-09; §2.3, §6) brought the records into line with it, corrected computed values (the field at the null with the beads; the beads as drawn), added first cuts, which stay PROPOSED until the designer accepts them, and ran four model checks (§6 items 31–34). Those correct both pumps' numbers: the record's values stay as its basis, with the corrected ones beside them, dated, until the designer decides (§5.2) |
| what the lock means | the design of record below is the baseline; later changes are recorded against it. The lock does not settle what is still PROPOSED or OPEN (§5): those keep their status |
| the redesign (2026-10-10) | the designer is redesigning the electrostatic pump's multipliers: stacked into the doubler, not a separate circuit tapping nodes 1 and 4. The turbine's stack will change; the magnetic pump, the sphere, its AH and the field interface stay for now. The worked circuit is §3.9, taken by the designer "for now" (2026-10-11); its mechanical layout is §3.9 too (PROPOSED). The record below stays the baseline until the designer accepts them |
| this ledger | `docs/ledger/DCCREG-design-ledger.md` (the source) and `.pdf` (its print form) |
| the drawings | `docs/ledger/DCCREG-drawings-bundle.pdf`: 60 sheets behind a two-page register (§7) |
| how it is built | `python3 docs/ledger/make_ledger.py` (needs markdown-it-py, mdit-py-plugins, pypdf and playwright with Chromium) |

**Abstract.** As locked, the DCCREG turbine is a belt-driven tube machine. Its rotor and counter-rotor turn at 600 rpm in opposite directions through a 1 : −1 reversing gear.

- **Two pumps on each shaft half,** both run by the relative rotation at 120 Hz:
  - **the magnetic pump:** three wound utrons pass six iron bridges, wired as the planar dual of a diode charge doubler, and drive the anti-Helmholtz (AH) coil pair;
  - **the electrostatic pump:** an air vane stack in a de Queiroz diode doubler drives two copper rings on the outside of a 50 mm glass vacuum sphere, through two mirrored Cockcroft-Walton chains.
- **Two static fields at the sphere's centre:**
  - the AH's steady magnetic cusp (300 ampere-turns per coil as recorded, about 250–270 in the best estimate, §3.3), with its null at the centre;
  - a DC electric field of 7.62 kV/cm (2.57 Pa) at switch-on, settling to 8.2 kV/cm as the insulators leak. It points from ring B to ring A and is steady to 0.19 %. As built, with the tube's strays and real HV diodes solved, it is about 6.8 kV/cm, and the pump needs a seed to start (§3.4).

- **Where the power goes:** the pumps take about 21.5 W from the belt as recorded (about 15–17 W in the best estimate), and the windage and the bearings about 17 W more: about 39 W in all (32–34 W). All of it ends as heat in the clamps, the copper, the iron, the air and the bearings. The products are the fields.
- **What the model checks found:** both pumps are weaker than recorded.
  - The magnetic pump's early gain falls to 1.03–1.05 with the utrons in 3-D and the neck's field map, and its kick must fire at full speed.
  - The electrostatic pump as built, its vanes' rims and the tube's strays solved, gains only 1.03 a cycle with ideal diodes; with real ones it does not self-excite.
  - The remedies are the designer's (§5.2).
- **The redesign (2026-10-10, §3.9):** the designer is stacking the multipliers into the doubler.
  - The worked proposal (PROPOSED) runs the pump bipolar, with C1 / C2 in phase, and continues each chain one stage past its varicap node to its ring.
  - As built it holds the rings at ±14.93 kV (±14.67 kV with typical HV sticks; the record ±13.86 / ±13.39 kV) on 12 single sticks, at a lower ideal gain (1.014 against 1.032).
  - **The layout (2026-10-11, PROPOSED):**
    - Ca / Cb stay plates in each side's column, toward the hub;
    - every diode and the multiplier's capacitors sit on a potted deck round a grounded can over the hub;
    - only l1 and r1 cross between the halves; the machine grows to 974 mm, and to 1014 mm with its bridge-ring bearings.
  - **What it costs:**
    - it needs the bridge-ring bearings, since the hub-face bearings must go. They are placed and named: an SKF 6205-2Z at each bridge ring's outer end, 83–87 Hz at d 25 (PROPOSED, §3.9);
    - it needs the clamp strings trimmed to about 60 parts.
- **What is still open:** the physics is mainstream throughout [OC]. The ratings, the leakage and several parts are placeholders that the bench test must qualify.

Generated from docs/ledger/DCCREG-design-ledger.md by docs/ledger/make_ledger.py, built at commit bbaf186. The markdown is the source; this PDF is its print form.

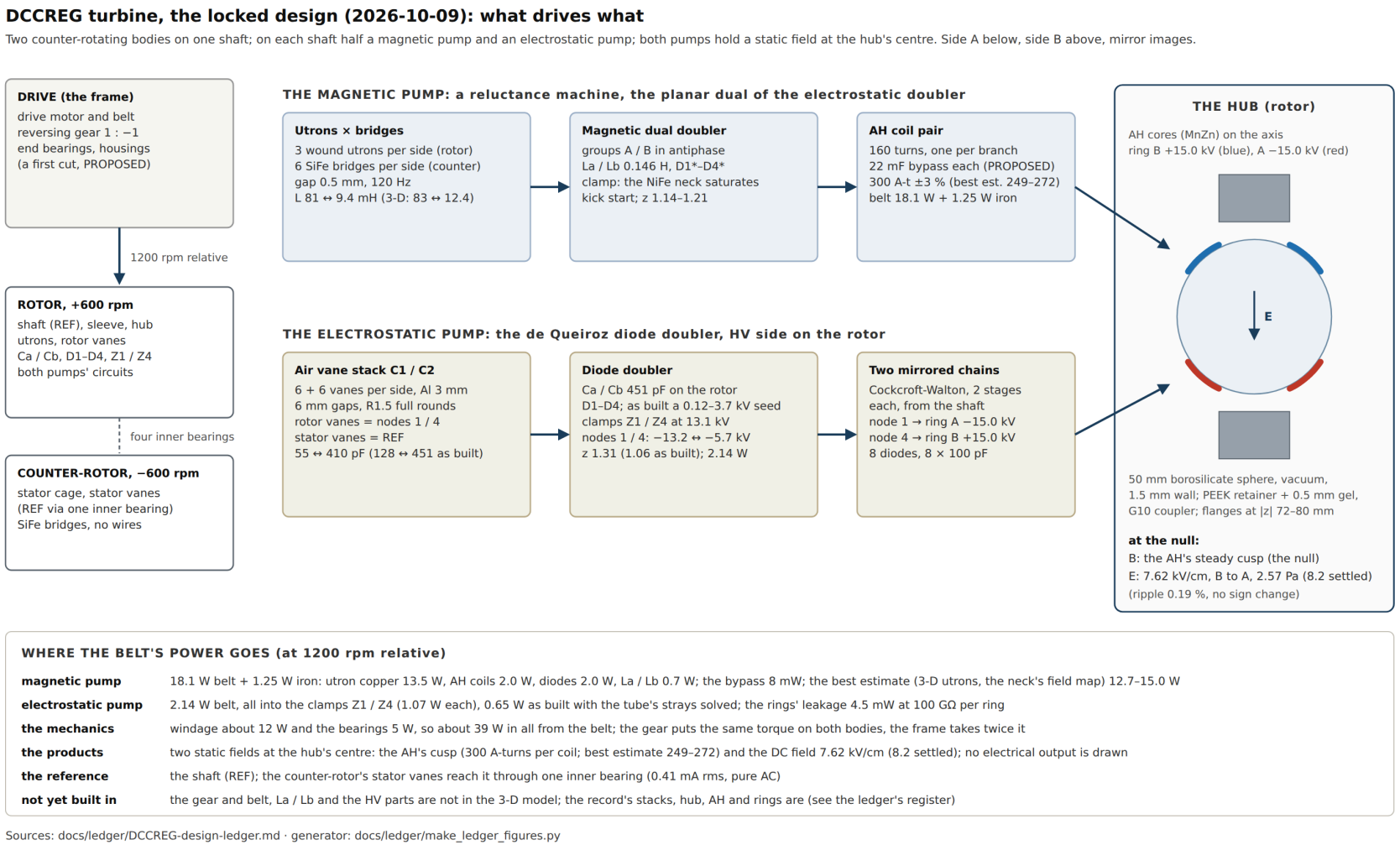
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**Reading notes.**

- **Tags** (`CONVENTIONS.md` §1):
  - [OC] standard, derivable physics;
  - [IR] a modelling or engineering choice, including datasheet-class values;
  - [RH] a heuristic or placeholder, not load-bearing until qualified.
- **Status words** (as the presets use them):
  - DECIDED: by the designer;
  - PROPOSED: worked out here and used, not yet accepted;
  - REGISTER: carried over from an earlier register;
  - DESIGN: the current sizing;
  - OPEN;
  - ESTIMATE.
- **"The record"** is the design of record.
- **Sources:** every number cites the file that holds it; the paths are from the repository root.

# 1. The machine at a glance




**WHERE THE BELT'S POWER GOES (at 1200 rpm relative)**

| | |
|---|---|
| **magnetic pump** | 18.1 W belt + 1.25 W iron: utron copper 13.5 W, AH coils 2.0 W, diodes 2.0 W, La / Lb 0.7 W; the bypass 8 mW; the best estimate (3-D utrons, the neck's field map) 12.7–15.0 W |
| **electrostatic pump** | 2.14 W belt, all into the clamps Z1 / Z4 (1.07 W each), 0.65 W as built with the tube's strays solved; the rings' leakage 4.5 mW at 100 GΩ per ring |
| **the mechanics** | windage about 12 W and the bearings 5 W, so about 39 W in all from the belt; the gear puts the same torque on both bodies, the frame takes twice it |
| **the products** | two static fields at the hub's centre: the AH's cusp (300 A-turns per coil; best estimate 249–272) and the DC field 7.62 kV/cm (8.2 settled); no electrical output is drawn |
| **the reference** | the shaft (REF); the counter-rotor's stator vanes reach it through one inner bearing (0.41 mA rms, pure AC) |
| **not yet built in** | the gear and belt, La / Lb and the HV parts are not in the 3-D model; the record's stacks, hub, AH and rings are (see the ledger's register) |

Sources: docs/ledger/DCCREG-design-ledger.md · generator: docs/ledger/make_ledger_figures.py

Figure 1. The locked machine: what drives what. The drive turns the two bodies in opposite directions; on each shaft half a magnetic pump and an electrostatic pump run from the relative rotation; both hold a static field at the hub's centre. Source: docs/ledger/figures/architecture.svg (docs/ledger/make_ledger_figures.py).

**The bodies.**

- **The rotor** carries:
  - the split shaft, which is the electrical reference (REF);
  - the hub;
  - the rotor vanes of the electrostatic stacks;
  - the wound utrons;
  - every circuit part of both pumps.
- **The counter-rotor** carries:
  - the stator cage;
  - the stator vanes, which are REF for the varicaps;
  - the passive iron bridges. It holds no wires, and it reaches the shaft through one inner bearing, the reference link.
- **The frame** holds the two end bearings, the reversing gear and the drive. The gear puts the same torque on both bodies, and its carrier, held by the frame, takes twice it (`docs/drive-gear-belt.md` §1).
- **The axis** is vertical: side A below, side B above, mirror images about the hub's equator.

**The two pumps** run at 1200 rpm relative, 120 Hz.

- **The magnetic pump:**
  - each side's three utrons form one group, A or B, whose inductance swings 9.4 ↔ 81 mH per coil as the bridges pass (12.4 ↔ 83 mH in 3-D, `sim/utron-3d-findings.md`);
  - groups A and B run in antiphase, in a circuit that is the exact dual of a diode charge doubler;
  - the saturating NiFe neck in each utron is the clamp;
  - one AH coil sits in each group's branch, and a 22 mF bypass across it (PROPOSED) holds a steady 300 ampere-turns (221–245 with the 3-D utrons; 249–272 with the neck's field map as well, the best estimate).
- **The electrostatic pump:**
  - the varicaps C1 / C2 swing 55 ↔ 410 pF in antiphase (128 ↔ 451 pF as built, with the vanes' rims, `sim/tube-strays-findings.md`);
  - the de Queiroz diode doubler multiplies its charge each cycle until the clamps Z1 / Z4 hold the nodes at their 13.1 kV operating peak;
  - the pump's nodes 1 and 4 then feed two mirrored Cockcroft-Walton chains standing on the shaft, which pump ring A to −15.0 kV and ring B to +15.0 kV.

**The hub** is a 50 mm borosilicate vacuum sphere with a 1.5 mm wall.

- The two MnZn AH cores sit on its axis.
- The two copper-foil rings lie on the glass, under 0.5 mm of silicone gel in a PEEK retainer, with a G10 coupler outside.
- At its centre the AH's null and the rings' field coincide.

**The redesign in progress** (2026-10-10, §3.9).

- The designer has decided to stack the charge-pump multipliers into the doubler.
- The proposal (PROPOSED) runs the pump bipolar, with C1 / C2 in phase, and feeds each ring from one stage past its varicap node.
- The numbers below are the record's.

**The design in numbers**

| quantity | value | source |
|---|---|---|
| speed | 600 rpm each way, 1200 rpm relative; both pumps at 120 Hz | `sim/pole-design-findings.md` §6; `sim/core_field.py` |
| tube | 940 mm: the air stack of record and the locked hub in solids, 153 solids without a clash | `sim/tube-shaft-findings.md` §0 |

| quantity | value | source |
|---|---|---|
| diameters | vanes Ø300 mm; bridge ring Ø313 mm; stator cage Ø332 mm | `sim/pole-design-findings.md` §8 |
| magnetic pump, per side | 3 wound utrons (200 turns, 2.23 kg each) and 6 SiFe bridges, gap 0.5 mm | `sim/pole-design-findings.md` §8 |
| AH | 160-turn coils on Fair-Rite 77 MnZn rods (0.80 mm wire, 0.333 Ω as wound); 22 mF bypass each (PROPOSED); 290–308 A-turns (300 mean, ±3 %; 294 as wound) with the record's 2-D utrons, 221–245 with the 3-D ones | `sim/ah-steady-cusp-findings.md`; `sim/hub-thermal-findings.md` §1; `sim/utron-3d-findings.md` §4 |
| the AH's null | at the centre; 0.113 T/m (0.083–0.092 with the 3-D utrons, 0.094–0.102 with the neck's field map as well); within ±0.44 mm over a cycle with the bypass, ±6.4 mm without | `sim/ah-null-findings.md`; `sim/utron-3d-findings.md` |
| electrostatic pump, per side | 6 + 6 Al vanes 3 mm, full rounds R1.5, 6 mm air gaps; C1 / C2 55–410 pF; Ca / Cb 451 pF. As built (a 3-D field solve): C1 / C2 128–451 pF (κ 3.51), Ca 480.5 pF, node strays 66 / 25 pF; z 1.058 bare, 1.031 with the rings' chains | `sim/air-stack-sizing-findings.md` §6.5; `sim/tube-strays-findings.md` |
| operating peak | 13.1 kV at the clamps (the 6 mm gap's breakdown over 1.5) | `sim/air_stack_sizing_results.json` |
| rings | Cu foil 0.10 mm, 26.25–55.71° from the axis; beads Ø3 / Ø2 mm; 29.9 mm apart along the glass | `docs/rings-design.md` §2 |
| rings' supply | symmetric: −14.96 / +14.96 kV on 2 + 2 Cockcroft-Walton stages from the shaft (−14.65 / +14.67 kV with real HV sticks; as built −13.38 / +13.40 kV) | `sim/hub_rings_build_results.json` record; `sim/diodes-real-findings.md` §4.3, §6.5 |
| field at the null | 7.62 kV/cm from B to A at switch-on, 2.57 Pa (the bands alone 7.10); 8.20 kV/cm once settled at 25 °C; ripple 0.014 kV/cm p-p (0.19 %); no sign change in a revolution. With real HV sticks 7.47 kV/cm at switch-on and 0.049 kV/cm p-p (6.96 kV/cm at their datasheet's maximum leakage). With the tube's strays solved, 7.06 kV/cm (±13.86 kV); as built with real HV sticks 6.82 kV/cm, 6.31 at their maximum leakage, and hot the pump does not start from the deck's −1 kV | `sim/hub-beads-settled-findings.md` §2; `sim/hub_drift_results.json`; `sim/hub_revolution_results.json`; `sim/diodes-real-findings.md` §4.3, §6.5; `sim/tube-strays-findings.md` §5 |
| power from the belt | magnetic 18.1 W + 1.25 W iron (10.2–12.3 W with the 3-D utrons, 12.7–15.0 W with the neck's field map as well, + iron); electrostatic 2.14 W (0.65 W as built); windage about 12 W and bearings 5 W | `sim/ah-steady-cusp-findings.md`; `sim/utron-3d-findings.md`; `sim/core_field_results.json`; `sim/tube-strays-findings.md`; `sim/rotor-mechanics-findings.md` §5 |
| the hub's heat | the AH coils 1.22 W each: the coils at 43 °C, the glass at 32–37 °C in a 25 °C room | `sim/hub-thermal-findings.md` |
| cost (placeholders) | the stack of record's build 6,477 EUR; the cheapest qualifying build 6,060 EUR | `docs/cost/README.md` |

# 2. How the design got here

## 2.1 What the machine is for, as the record states it

- **The brief now,** in the designer's words (`sim/core-null-field-findings.md`): the centre of the vacuum core, where the AH null sits, should carry the strongest electrostatic field (pressure).
- **The AH pair** should hold a steady cusp (`sim/ah-steady-cusp-findings.md`).
- **Both pumps** are the supplies, driven from one belt through the 1 : −1 gear (`sim/hub-drive-findings.md`).
- **What the field at a magnetic null is to be used for is not documented.** This ledger records the brief as given and does not supply a purpose.

## 2.2 The path, in eleven steps

Each step asked a question, found an answer in mainstream physics and left a decision behind. The dates are the commits' dates.

| # | step (dates) | the question | what was found | what it left |
|---|---|---|---|---|
| 1 | the browser tool (to 06-09) | a sectored-disc varicap driving a symmetric Bennet doubler | blocks C-I, M, R, D, T, S in `index.html`; anchor z 1.203 | the tool, now retired and frozen |
| 2 | the disc spark-gap machine (06-11 → 06-29) | can a doubler with spark-gap commutation, a resonant tank and islands deliver energy | the doubler's equalisation *is* the pump; only a downstream sink recovers energy; η ≈ 0.70 is forbidden, η ≈ 0.45–0.50 (`docs/efficiency-resolution.md`) | the freeze v0.10 (C_R 789 pF); the efficiency principle |
| 3 | the drawn pump re-checked (09-29 → 10-02) | does the designer's drawn pump pump, exactly | exact engines and gates (PUMP-CALC, PUMP-SYNTH, GEOM stage 2, the integrity tool); z 1.325 on the drawing | the tools, frozen |
| 4 | the round trip and the design loop (10-03 → 10-05) | do the solids keep the gain | the field-solved floors and strays (about 1 nF) take z to 0.83–0.89 (NO-FLOOR-IN-RANGE); interleaved stacks recover 1.24; with the motor in, z = 1.000 | interleaved stacks of vanes are the way to C; the motor cannot sit across Ca |
| 5 | the register and the motor (10-05) | can the C-EMs turn the counter-rotor | they are about five orders short, and the pump's own reaction torque (78 mN·m) needs an outside drive | the belt and the reversing gear |
| 6 | the tube (10-05 → 10-07) | an elongated machine with vane stacks along the shaft | C1 1.11 nF, κ 15.7 in vacuum; diodes beat spark gaps (η 1.00, 3.0 W against 1.3 W at η 0.53) | the de Queiroz diode core is the electrostatic pump |
| 7 | the pivot: two geared pumps (10-07 → 10-08) | drive the AH from the belt too | the planar dual of the doubler works (z 1.528 against 1.512); wound utrons and passive bridges; the mean-turn fix moves the pick to 2.34 kg, 200 turns, 18.8 W | the magnetic pump of record |
| 8 | the air build (10-08) | the first tests run in air | air takes about a third of the vacuum field; 6 mm gaps, full-round vanes, the 6 + 6 cap: 2.14 W; the AH bypass for a steady cusp; the vane matrix and the cost sheet | the electrostatic stack of record |
| 9 | the field at the core (10-08) | the strongest field at the AH null | HV side onto the rotor (unchanged pump); cones steady or swinging (≈1.4 kV/cm); a DC pair inside the vacuum (65.7 kV/cm, but feedthroughs) | the hub locked; the field needs electrodes |
| 10 | the rings (10-09) | rings outside the glass | the beads, not the separation, set the stages; asymmetric 6.93 kV/cm → symmetric 7.10 kV/cm on the bands alone; steady, no swing | the rings and supply of record (`docs/rings-design.md`) |
| 11 | the settlement (10-09) | do the records agree, and what can be settled before the bench | the lock's 18 inconsistencies resolved and 12 more found and settled; with its beads the null reads 7.62 kV/cm; the beads settled, the hub's heat, the AH null, the rotating mechanics, the drive and the parts' first cuts, the model checks (§6) | the decisions of §5.2 |

## 2.3 The designer's decisions

| date | decision | where recorded |
|---|---|---|
| 06-16 | the freeze v0.10 of the disc machine (12 mm septum, C_R 789 pF) | `docs/varcap-design-freeze-v0.10.md` |
| 10-03 → 10-05 | the design loop's rulings: the machine floats; septum ≤ 1000 mm; N = 2; garolite, not PTFE | `sim/rt-design-predictions.md` |
| 2026-10 (pivot) | rotor and counter-rotor geared 1 : −1; the AH on a magnetic pump; both pumps on the belt | `sim/hub-drive-findings.md` |
| 10-08 | "the first tests run in air"; vanes with full-round edges; the 6 + 6 cap | `sim/air-stack-sizing-findings.md` |
| 10-08 | "the AH pair should hold a steady cusp" | `sim/ah-steady-cusp-findings.md` |
| 10-08 | the reference link through an inner bearing "for now", a brush later | `sim/rotor-parts-duty-findings.md` §4 |
| 10-08 | the HV side on the rotor | `sim/core-field-findings.md` (decision 1) |
| 10-08 | the hub locked: layer order, the 50 mm borosilicate sphere, symmetric pumps | `presets/hub-locked.json` |
| 10-09 | the rings outside the glass; a 1.5 mm wall; the AH flanges outside the vessel | `presets/hub-locked.json`; `sim/hub-locked-findings.md` |

| date | decision | where recorded |
|---|---|---|
| 10-09 | copper foil with rounded edges for the prototype, a fired-on coating later | `sim/hub-rings-build-findings.md` (brief) |
| 10-09 | "go multistage as long as the rings' separation can hold it" | `sim/hub-rings-build-findings.md` (brief) |
| 10-09 | the symmetric supply: "if it pumps against the core centre, yes!" | `docs/rings-design.md` |
| 10-09 | **"Let's lock the design for now."** | this ledger |
| 10-09 (the settlement) | keep the stack of record (6 mm gaps, V_op = breakdown / 1.5, 22° / 22°, r 150, Ca = 1.1 C_max) and the clamps Z1 / Z4 | §5.1 |
| 10-09 (the settlement) | accept the magnetic pick and the retainer's materials (PEEK, the silicone gel, the G10 coupler outside); the 22 mF bypass and the equatorial split stay PROPOSED | `presets/hub-locked.json`; §5.1 |
| 10-09 (the settlement) | rewrite `README.md` and `CLAUDE.md` for the locked machine | `README.md`, `CLAUDE.md` |
| 10-10 | redesign the charge-pump multipliers so they stack into the doubler, not "a separate circuit tapping on node 1 and 4"; the design stack of the whole turbine will change; the magnetic pump, the sphere, its AH and the field interface untouched for now | §3.9 |
| 10-10 | the sketch: C3–C6 and D5–D8 are the new parts (read back and confirmed); the clamps Z1 / Z4 stay on nodes 1 / 4; C3–C6 to be sized; the polarity free; the rings' supply to be worked out; the doubler, the rings and the magnetic pump's Schottky circuit on one revolving body | §3.9; `sim/integrated-multiplier-findings.md` |
| 10-11 | "This unipolar setup will do for now": the proposal's circuit, for now (read here as Figure 10's: each chain one polarity, ring A's side negative). "Lay out the doubler capacitors in the multiplier top and bottom of the machine stack. Propose placement of the diodes. A mechanical layout." | §3.9; `sim/multiplier-layout-findings.md` |

## 2.4 What was learned, and what it superseded

- **Charge pumps pay their equalisation tax inside the core** [OC].
  - A doubler with lossy transfers cannot recover what it spends, and only a downstream sink recovers energy. The disc machine's claimed η ≈ 0.70 is forbidden (`docs/efficiency-resolution.md`).
  - With smoothly varying capacitance and diodes the transfer itself is lossless (η 1.00), and that is the pump kept (`sim/diode-stack-findings.md`).
  - Spark gaps, islands, shuttles and the resonant tank are superseded.
- **Strays and floors set the gain, not size** [OC].
  - The disc's field-solved floors and strays scaled with the build (`sim/round-trip-findings.md` §5).
  - Interleaved vane stacks along the shaft give the capacitance without the carriers' penalty, hence the tube.
  - In air the fixed 20 pF node stray is what pulls the capped stack's z from 1.38 to 1.31 (`sim/air-stack-sizing-findings.md` §6.5).
  - Solved in 3-D (`sim/tube-strays-findings.md`), the strays are 66 / 25 pF a node, and the vanes' rims, which the 2-D cell took as a 2 pF floor, add 65 pF to C_min: as built κ is 3.5, not 7.5, and z is 1.058, not 1.31.
- **A pump's reaction torque must go somewhere** [OC]. The C-EMs could not counter-rotate the stator, so the drive is an outside belt with a $1:-1$ gear, and every reaction torque goes into the frame (`sim/spinup-findings.md`).
- **The dual circuit works.** Inductors for capacitors, currents for voltages and flux for charge give a magnetic pump with its own gain. A saturating neck is its clamp, which makes it the natural driver for the AH coils (`sim/magnetic_doubler.py`, `sim/pole-design-findings.md`).
- **Air sets the voltage, the rims the vanes and the cap the power** (`sim/air-stack-sizing-findings.md`):
  - 6 mm gaps for a 13.1 kV peak;
  - 3 mm full-round vanes for the corona margin;
  - 6 + 6 vanes for 2.14 W.

- **The field at a null needs electrodes; outside the glass they need edges** (`sim/hub-rings-build-findings.md`).
  - The vacuum pair inside reached 65.7 kV/cm but needed two HV feedthroughs.
  - The rings need none. Their beads facing the AH coil ends, not their separation, set how many stages they hold.
- **Solve what is built** [IR]. The beads were sized in local boxes and never put back in the hub, so the record's null missed their 7 %; the drawing's grooves bring the PEEK to the beads, which the boxes had filled with gel (`sim/hub-beads-settled-findings.md`).
- **An end correction is a model, not a measurement** [IR]. The utrons' 2-D section took the stack's ends as [RH] corrections, +3 % aligned and +30 % unaligned. In 3-D they are +5.9 % and +71 %, and a group's series coils couple round the machine, which the 2-D walls exclude: κ 6.7 (5.96–6.44 coupled), not 8.6, and the pick's gain falls under its own selection rule (`sim/utron-3d-findings.md`).
- **Only the difference between the rings makes field at the null** [OC].
  - The pump's nodes are twins, so identical circuits give none.
  - The record's mirror chains put the null at the shaft's potential and each ring 15 kV from it, and the field is DC (`sim/hub_revolution.py`).

# 3. The components and how they work

## 3.1 Bodies, drive, bearings and the reference

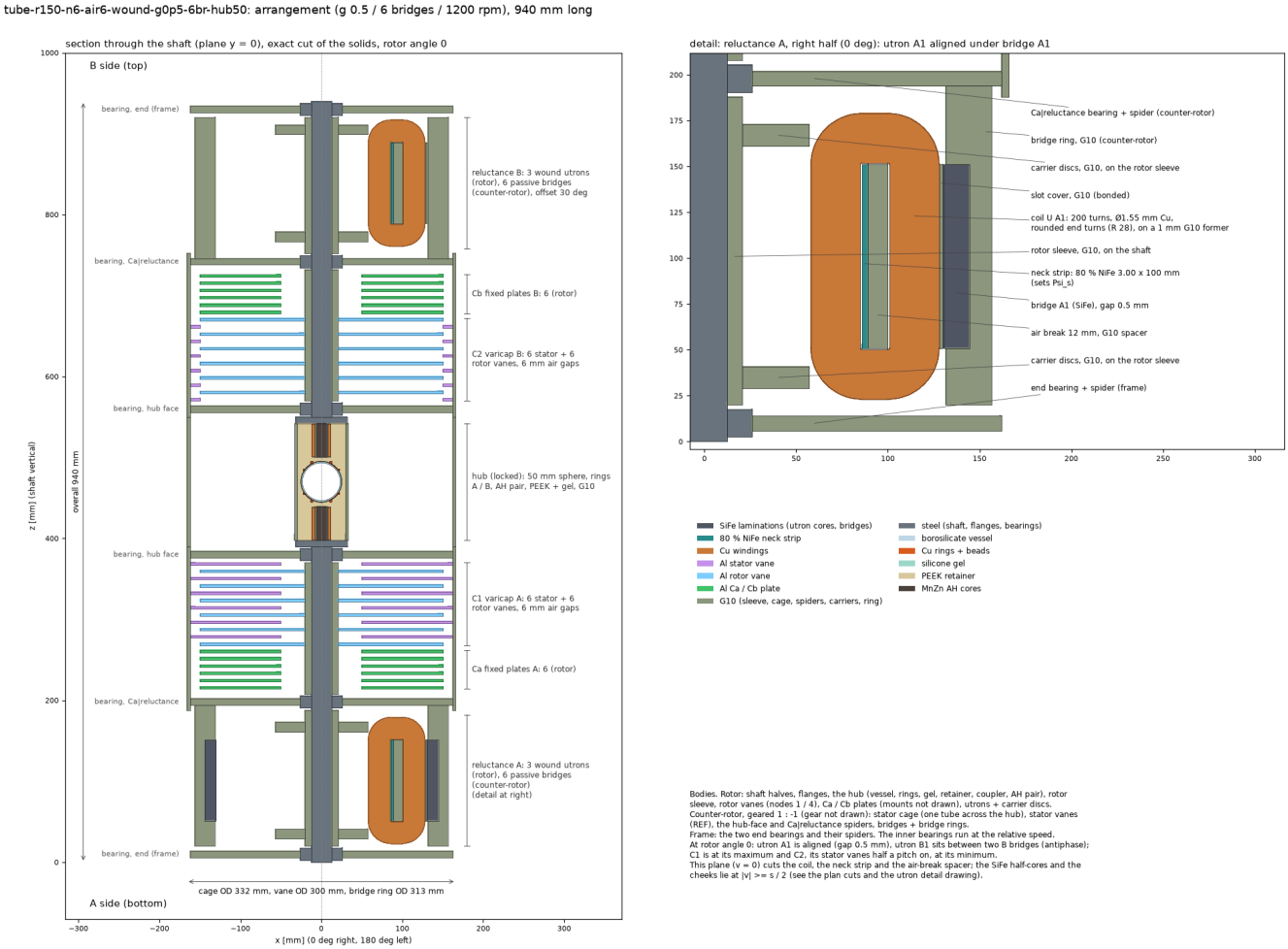



Figure 2. The machine of record in 3-D: the section through the shaft (side A below), with the air stack of record and Ca / Cb on the rotor, the reluctance sections and the locked hub; 940 mm. Source: docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-section.png (sim/tube_geometry.py --record).

- **The rotor** turns at +600 rpm and carries all of both pumps' circuits, so no wire crosses between the bodies (`sim/core-field-findings.md` §1, §7). It is the split shaft (two halves, each ending in a flange at the hub) with:
  - the G10 sleeve;
  - the rotor vanes (nodes 1 / 4);
  - Ca / Cb, D1–D4 and Z1 / Z4;
  - the utrons on two 12 mm G10 carrier discs per side;
  - La / Lb and D1*–D4*;
  - the hub.
- **The counter-rotor** turns at −600 rpm and carries no HV (`sim/tube_geometry.py`; `sim/pole-design-findings.md` §8). It carries:
  - the G10 stator cage (r 162–166 mm), one tube across the hub;
  - the stator vanes (REF);
  - the four inner bearing spiders;
  - six M235-35A bridges per side in a G10 ring, with side B's offset 30°.

- **The frame** holds the two end bearings and their housings, the 1 : −1 reversing gear, the drive motor and the belt. **The drive's first cut** (PROPOSED, `docs/drive-gear-belt.md`):
  - a bevel reverser at side A, its carrier held by the frame: two stainless side gears (z 30, m 2) and three POM-C pinions;
  - an HTD 5M belt, 4 : 1, from a 200 W brushless motor whose driver ramps and limits the current;
  - the gear puts 0.48 / 0.52 N·m on the bodies and 1.00 N·m into the frame; the belt carries 59 W with a 25 W windage allowance; the run-up takes 32 s;
  - the pumps' 120 / 240 Hz pulsation stays in the bodies: the gear sees 0.061 N·m p-p;
  - 1,066 EUR against the cost sheet's 800.
- **The bearings:** six 6205-class deep-groove bearings (25 / 52 / 15 mm), each on an 8 mm G10 spider.
  - The four inner ones join rotor and counter-rotor at the relative speed; only the two end bearings face the frame.
  - **So the counter-rotor floats on the shaft** (`sim/tube-shaft-findings.md` §0; `sim/rotor-mechanics-findings.md` §4, an independent model that agrees). As laid out (d 25) the first mode is 27–29 Hz, and 1 g lateral moves the utron gap 123 µm at the stack's centre and 1.7× that at its outer end.
  - **The ways out:** d 40 gives 65 Hz and 25 µm. A bearing between the shaft and each bridge ring's outer end gives 131–165 Hz (realistic to stiff seats) and 2–3 µm, but on G10 spiders it sits only 9 % above the pumps' 120 Hz, and the layout grows 10–20 mm a side for it. The choice is the designer's (§5.2). The redesign's layout needs the pair; there it is placed and named (an SKF 6205-2Z, 20 mm a side; §3.9).
  - **The gap budget:** each inner bearing's total runout within 0.027 mm as laid out (P5 bearings under an axial spring preload), or 0.05 mm with the eighth pair (`sim/rotor-mechanics-findings.md` §5).
- **The shaft:**
  - it is d 25 in the tube model and d 30 in the hub's register: OPEN, with the bearings (§5.2). A 6205 needs 25 mm journals;
  - non-magnetic, µ_r ≤ 1.05: annealed austenitic stainless. Steel halves would tie the AH null to how each seats, and titanium drops the first mode to 21 Hz (`sim/ah-null-findings.md` §5; `sim/rotor-mechanics-findings.md` §4);
  - it is split at the hub, each half bolted at its flange (r 32 × 8 mm at ±72–80 mm in the locked hub).
- **The rotating loads** (`sim/rotor-mechanics-findings.md`):
  - no banding is needed: at 750 rpm every part holds with a factor of 19 or more, if the studs are preloaded and the joints bonded in the impregnation;
  - balance each body to G2.5;
  - windage takes about 12 W and the bearings 5 W.
- **The reference link.** The counter-rotor's stator vanes are REF for C1 / C2. They reach the shaft through one inner bearing:
  - the path is rail → lead → outer ring → balls → inner ring → shaft;
  - it carries the varicaps' displacement current, 0.41 mA rms of pure AC with no DC (`sim/core_field_results.json`).
  - A floating counter-rotor would need about 10 nF to the shaft, and at 100 pF the pump stops, so the bearing link stays. It is not acceptable with a spark dump or without clamps, where the surges need a brush (`sim/rotor-parts-duty-findings.md` §4).

## 3.2 The magnetic pump: utrons, bridges and the dual doubler

**How a bridge makes an inductance that swings** [OC].

- A utron is a U-core electromagnet on the rotor. Its two radial tips face the counter-rotor at r 130 mm.

- When a passive SiFe bridge spans both tips, the flux closes across two 0.5 mm gaps and the coil's inductance is high: 81.0 mH. Between bridges the nearest iron is 9.74 mm away and it falls to 9.4 mH (κ 8.6).
- **In 3-D** (`sim/utron-3d-findings.md`, model check 34) the stack's ends add +71 % unaligned against the record's [RH] +30 %: 83.3 / 12.4 mH, κ 6.70. The group's three series coils also couple round the machine, which the 2-D section excludes: κ 5.96 with the coils aiding, 6.44 with one of the three reversed. How the three are connected is not recorded: the designer's.
- Six bridges on a 60° pitch at 1200 rpm relative swing it at 120 Hz. Each bridge sees the three utrons pass, so it carries unipolar flux at 60 Hz (`sim/pole-design-findings.md` §8).

**The dual doubler** (`sim/magnetic_doubler.py`).

- The electrostatic diode doubler's circuit graph has a planar dual: capacitors become inductors, voltages currents, charge flux.
- In it, the two varicaps become the two utron groups L1(θ) / L2(θ), three utrons in series each (0.243 H). Ca / Cb become the chokes La / Lb (0.146 H), and the diodes become D1*–D4*.
- The bridges of side B sit half a pitch on from side A's, so the groups are in antiphase.
- When a group's inductance falls while its flux is held, its current rises: it generates, and the belt does the work against the bridges' pull. When the inductance rises the group motors. The groups swap every half cycle.
- Per cycle the stored flux grows by a fixed factor: z 1.21 in the design screen's linear model, against a selection rule of ≥ 1.20. The loaded transient run reads 1.139 early on, and 1.147 with the AH bypass (`sim/pole_design_variants_op.json`; `sim/ah-steady-cusp-findings.md`).
- **With the 3-D utrons** the pick reads z_lin 1.16–1.17, under its own rule, and z_early 1.07–1.09. The AH gets 221–245 A-turns with the bypass instead of 300 (249–272 with the neck's field map as well, `sim/neck-nonlinear-findings.md` §6), and the belt pays 10–12 W (12.7–15.0 W) instead of 18 (`sim/utron-3d-findings.md` §4). Restoring 450 A-turns at the peak takes Ψs × 1.27–1.42 (a neck of about 3.8–4.3 mm), or a longer stack: the designer's (§5.2).
- Its duality check against the electrostatic doubler gave z 1.528 against 1.512 (`sim/hub-drive-findings.md`).

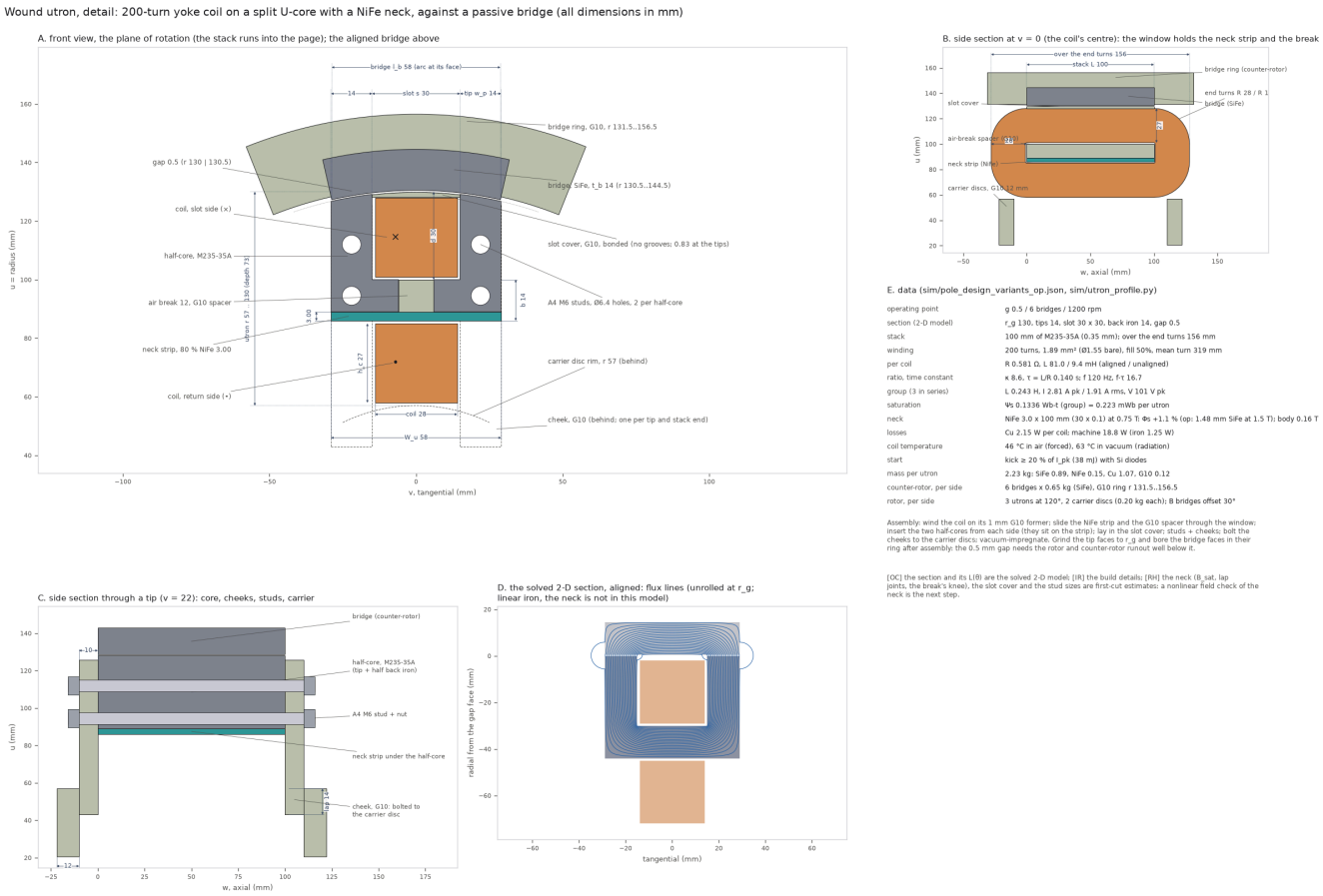



Figure 3. The wound utron against a bridge: the 200-turn coil on the split U-core, the 3.0 mm NiFe neck, the bonded slot cover, the studs; the solved 2-D flux. Source: docs/figures/utron-core-detail.png (docs/make_utron_drawing.py). The half-core's manufacturing drawing is DCCREG-UTR-101 (bundle sheet 4).

### The clamp, the start and the parts.

- **The clamp.** The dual of the avalanche clamps is the iron's saturation. A 3.0 mm 80 % NiFe strip under each utron's back iron saturates at Ψs 0.134 Wb-turns per group, and the current then stops growing. The law $i = \Psi/L\cdot(1 + (\Psi/\Psi s)^6)$ is a fit [IR]. The neck is sized to the AH's need and no larger, because heat goes as $\Psi s^2$ (`sim/pole-design-findings.md` §3, §8).
  - **The neck in a nonlinear field solve** (`sim/neck-nonlinear-findings.md`, model check 31): the strip saturates at Ψs 0.1268 Wb-turns, 5 % below the design, because 30 foils of 0.1 mm stack at about 0.90 [RH], not 1. Its foils lie across the radius, so at the laps the flux crosses them: the aligned L falls 10 % below the knee (72.8 mH, κ 7.78, z_lin 1.187 in the 2-D frame). Past it the knee is sharper than the ^6 fit, so the pump drives the AH harder: 323 A-turns per coil with the bypass (+8 %) for 21.0 W on the belt. Foils stacked axially would keep the record's gain (322 A-turns, z_early 1.147); a 3.2–3.3 mm strip would restore Ψs: the designer's.
  - **With the 3-D utrons as well** (§6 there, the best estimate [IR]): 249–272 A-turns per coil with the bypass (by how each group's coils are connected), z_early 1.028–1.051, 12.7–15.0 W on the belt; the pull stays about 28 N per utron.

- **The start.** With the deck's diodes (0.54 V at 1 A, a 100 V Schottky's drop) a small seed does not grow, because the winding voltage while growing is only a few volts. A one-time kick starts it: the record's 20 % of Ψs seeds 0.110 A and 3.5 mJ (`sim/pole-design-findings.md` §7); the threshold is 16 %, 2.2 mJ, with the deck's diodes (`sim/parts-first-cut-findings.md` §3).
  - **With real rectifiers** (`sim/diodes-real-findings.md` §3) the threshold follows the drop: 17.2 % (16.4 % with the bypass) for the first-cut Schottky set, 20.8 % for a 200 V Schottky in all four, 25–28 % for silicon PN rectifiers, which the record's 20 % kick does not start. Running, the first-cut set costs the AH 300 → 295 A-turns and puts 2.42 W in the diodes; reverse recovery is negligible at 120 Hz.
  - **At full speed,** 600 rpm each way and never below 500: the pump starts at every rotor phase only from 1000 rpm relative, and a running pump stops below about 750. At 200 rpm each way the kick dies within 8 cycles.
  - **With the 3-D utrons** (`sim/start-3d-findings.md`): K4 still starts the pump at full speed at every phase, with every set. It starts at every phase only from 1050–1100 rpm relative, so never below 550 each way; a running pump stops below 850; and the seed threshold is 20–25 %, so the record's 20 % seed kick starts only the record's frame.
  - **With the neck's field map as well** (the best estimate, `sim/neck-nonlinear-findings.md` §7): K4 still starts it at full speed at every phase, but starts every phase only from 1150 rpm relative (the record's frame) or at full speed (the coils aiding), and the seed threshold is 25–30 %. So K4 fires at full speed, 600 rpm each way.
  - **The first cut (PROPOSED):** a 9 V lithium cell charges a 470 µF bipolar capacitor, and an SCR dumps it across La (+ to node a) when a reed switch on the rotor passes an outside magnet: 19 mJ, 0.60 A, 17 ms, once per charge.
  - **The sign is the diodes':** a seed of the other sign starts the pump in the deck's own sign, so polarised bypass parts go + to nodes d / b.
- **The parts:**
  - the utrons: M235-35A 0.35 mm laminations, stack 100 mm, 200 turns of Ø1.55 mm Cu, 0.58 Ω, 2.23 kg each as built;
  - La / Lb: 0.146 H DC chokes, 0.87–1.15 A, ≤ 110 V. First cut (PROPOSED): EI-84 laminations, a 35 mm stack, 150 turns of Ø1.40 mm, a 0.126 mm gap trimmed to 0.146 H; R20 0.276 Ω, 1.20 T at the peak, 1.66 kg each (`sim/parts-first-cut-findings.md` §1);
  - D1*–D4*: reverse 113 / 104 / 58 / 60 V with the start-up, 1.9 / 2.3 A peak in each diode (2.8 A in the branch). First cut (PROPOSED): Schottky, 200 V (D1* / D2*) and 150 V (D3* / D4*), ≥ 3 A. Silicon rectifiers (0.85 V) would cost z 1.139 → 1.074 and the AH 449 → 423 A-turns (§2.1 there);
  - snubbers: an RC at each of the eight nodes (21 nF + 243 Ω in the deck; first cut 22 nF film + 240 Ω) [IR];
  - the wiring stray Lp2 / Lp3: 4.4 mH [IR];
  - the coils run at 46 °C in air (63 °C in vacuum), at the model's 40 °C ambient (`sim/pole_design.py`).
- **The power:** 17.6 W on the belt without the bypass (utron copper 12.9, AH 2.00, La / Lb 0.66, diodes 2.01), plus 1.25 W of iron. With the bypass the belt reads 18.1 W (`docs/schematic-rotor-circuits.png`; `sim/ah-steady-cusp-findings.md`).

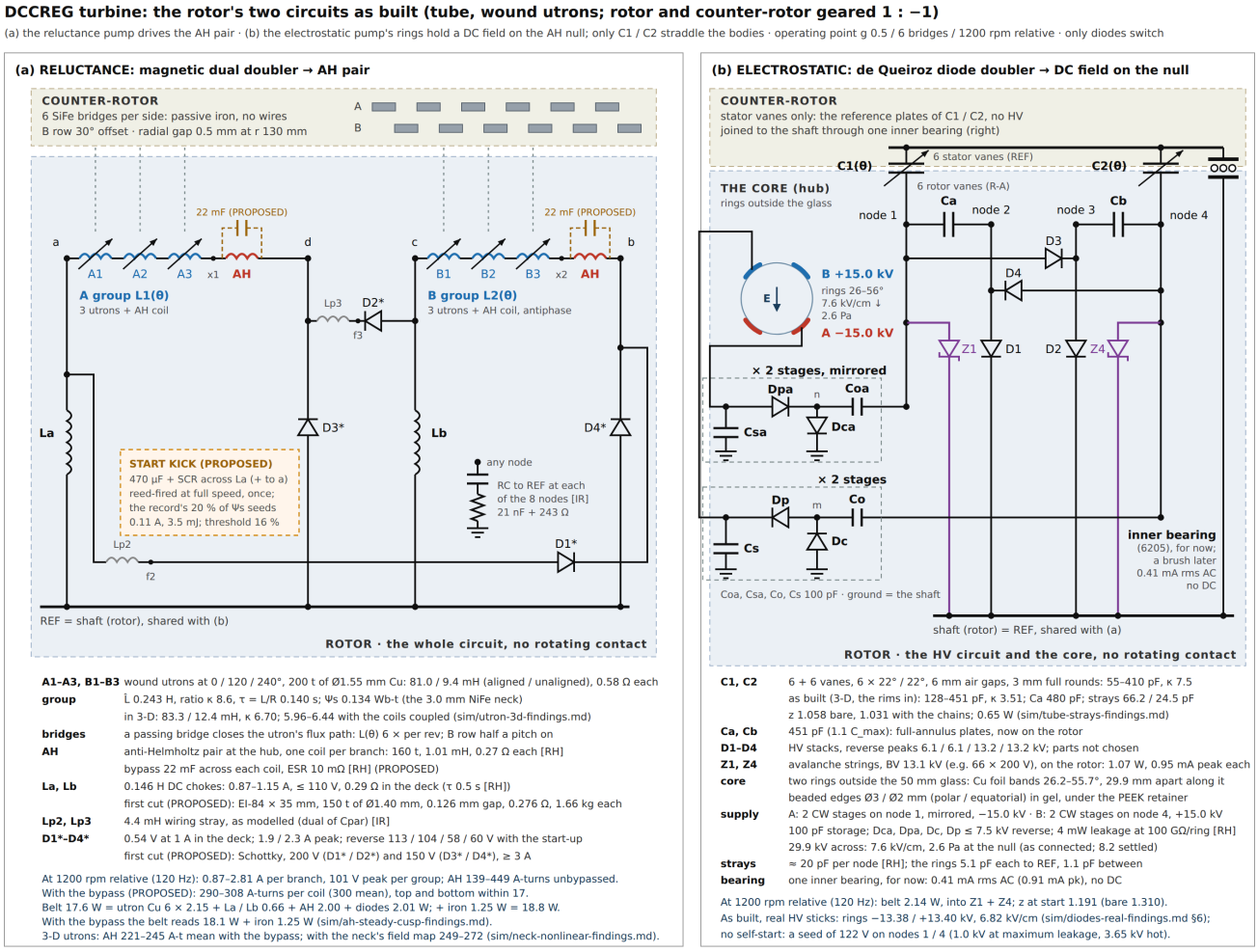



Figure 4. The rotor's two circuits as built: (a) the magnetic dual doubler driving the AH pair; (b) the electrostatic de Queiroz doubler with the HV side on the rotor and the rings' two mirrored chains. The 22 mF bypass across each AH coil is drawn dashed (PROPOSED). Source: docs/schematic-rotor-circuits.svg (docs/make_schematic_rotor.py).

## 3.3 The AH pair and the steady cusp

- **The pair:** two coils on the shaft's axis, one in each group's branch:
  - 160 turns each, 1.01 mH. As wound, 0.80 mm grade-1 wire in 4 orthocyclic layers fills the window: 0.333 Ω at 20 °C, against the deck's 0.27 Ω, which scaled a window of bare copper (`sim/hub-thermal-findings.md` §1). It drives 294 A-turns instead of 300 with the bypass;
  - wound on G-10 formers around Fair-Rite 4077484611 rods of 77 MnZn ferrite, Ø12.3 mm, at ±30.7–72 mm;
  - seated in PEEK 5.7 mm from the vessel (`presets/hub-locked.json` AH, AH_seat).
- **Anti-Helmholtz:** they are wound to oppose, so their fields cancel at the centre (the null) and rise linearly away from it, the cusp [OC].
  - **Each branch carries its own side's coil.** Both coils' currents are unipolar and of the same sign in their elements' orientation (x1 → d, x2 → b).
  - **How to wind them:** two identical coils, each with its start lead to its group's utrons and its finish to the diode side, mounted as rotated copies with their start leads toward the vessel. A translated copy would make the pair Helmholtz (`sim/ah-steady-cusp-findings.md`, the winding sense).

- **The null's position** (`sim/ah-null-findings.md`):
  - balanced, it sits at the centre, with 0.113 T/m along the axis (0.377 mT/m per A-turn on both coils; 0.111 T/m as wound), −½ of that across. The rods raise it 4.8×;
  - each A-turn of imbalance moves it 25 µm toward the weaker coil. With the bypass it stays within ±0.44 mm over the 120 Hz cycle; without it it swings ±6.4 mm, beyond the ±5 mm where the rings' field holds 4.4 %;
  - the earth's vertical field moves it 0.35–0.5 mm; a 0.5 mm misplaced coil, 0.25 mm.
- **The steady cusp.**
  - Without a bypass, each coil carries its branch's unipolar 139–449 A-turns at 120 Hz. Top and bottom peak in turn, so the null moves along the axis every cycle.
  - A 22 mF capacitor across each coil (34 Hz with the coil, far below 120 Hz) carries the ripple. Each coil then holds 290–308 A-turns (300 mean, ±3 %), and top and bottom stay within 17 A-turns (`sim/ah-steady-cusp-findings.md`).
  - 2.2 mF would resonate near 107 Hz and make the ripple worse. The capacitor's duty is light: 0.65 A rms, about 0.5 V, 8 mW for the pair.
- **The cost of steadiness:** the steady field is the mean, 300 A-turns, against the 449 A-turn peak the AH was sized to. A steady 450 needs 240 turns or a larger pump, which is not re-sized.
- **With the 3-D utrons** (§3.2) the pump settles lower: 237–251 A-turns with the bypass in the record's frame (245 mean), 229–242 with one coil of each group reversed and 214–227 with the coils aiding; so 0.083–0.092 T/m at the null (`sim/utron-3d-findings.md` §4).
- **The rods** peak at 0.121 T at 300 A-turns. They reach the register's 0.30 T limit [RH] at 743 A-turns for the AH alone; the record's "about 600" scaled a run with the cone windings in series (`sim/ah-null-findings.md` §4).

## 3.4 The electrostatic pump: the vane stack and the diode doubler

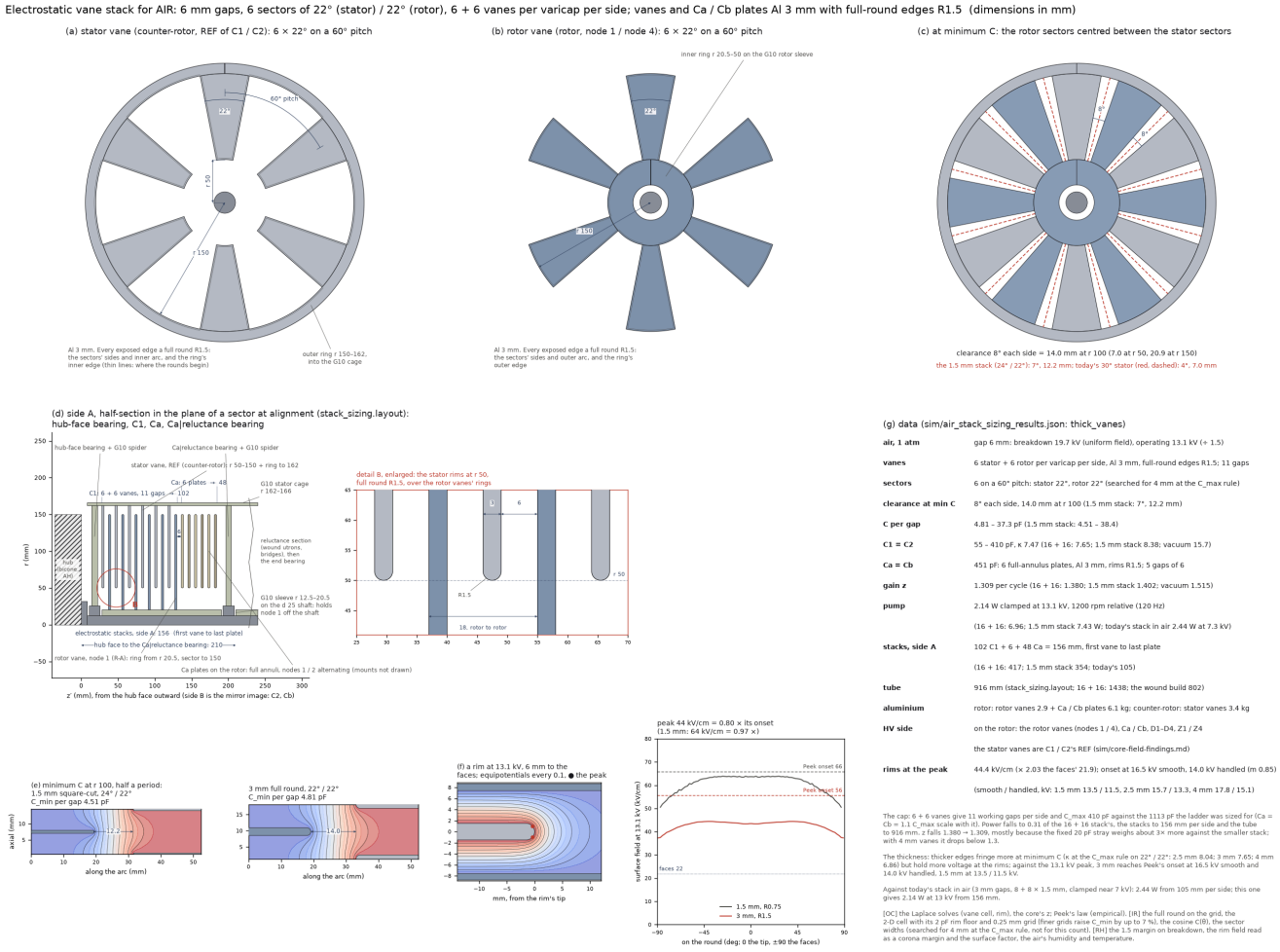



Figure 5. The air vane stack of record: the stator vane (REF), the rotor vane (node 1 / 4), both at minimum capacitance, side A's half-section, the vane cell and the rim field against Peek's onset. Panel (d) still draws the placeholder hub. Source: docs/figures/air-vane-stack-6mm.png (docs/make_air_vane_drawing.py).

**The varicaps C1 / C2** (`sim/air-stack-sizing-findings.md` §6.5).

- On each side, 6 stator vanes (REF, counter-rotor) interleave with 6 rotor vanes (node 1 on side A, node 4 on side B) across 11 gaps of 6 mm in air.
- The vanes are aluminium, 3 mm thick, every exposed edge a full round R1.5. Each has six 22° sectors on a 60° pitch, r 50–150 mm.
- With the sectors aligned the stack is 410 pF. With the rotor's sectors centred between the stator's it is 55 pF (κ 7.47).
  - **As built** (a 3-D field solve of the record's solids, `sim/tube-strays-findings.md` §4): 450.6 / 128.4 pF, κ 3.51. Each stator sector's inner rim sits over the rotor ring's edge, and each rotor sector's outer rim under the stator ring's, edge to edge 6 mm apart. The 2-D cell took these rims as a 2 pF floor on C_min; they add 65 pF to it and 37 pF to C_max [OC].
- C1 and C2 sit half a pitch apart, so they swing in antiphase six times a relative revolution: 120 Hz. (The record's solids drew them in phase; corrected, §6 item 58.)

**The doubler** [OC] (`sim/core_field.py`; de Queiroz's symmetrical generator, its Fig. 1 with every diode reversed: the same circuit at negative polarity).

- **The circuit:**
  - C1 from node 1 and C2 from node 4 to REF;
  - transfer capacitors Ca (1–2) and Cb (3–4) of 451 pF (480.5 pF as laid out: the layout rounds the gap count up);
  - four diodes: D1 2→0, D2 3→0, D3 1→3, D4 4→2.
- **One half cycle, step by step:**
  1. The capacitor that is shrinking keeps its charge, because its diode is blocked, so its voltage rises as Q/C. The rotation does work against the vanes' attraction.
  2. Through Ca or Cb that lift pulls the next node until a diode conducts.
  3. The charge then pours onto the other varicap while it is large, so it accepts the charge at low voltage.
- **The gain.** Charge is collected while a capacitor is large and lifted while it is small, so it grows each cycle by a fixed factor: z 1.31 bare (1.3095 in ngspice). With the rings' chains attached, z is 1.191 at start-up: a large-signal gain, which real HV sticks keep from about 1 kV up (1.14–1.19; `sim/diodes-real-findings.md` §4.1).
- **Self-excitation, with ideal diodes only.** Between diode events the ideal circuit is linear and scale-free, so any seed in the growing mode multiplies by z every cycle [OC].
  - **Corrected 2026-10-09** (`sim/diodes-real-findings.md` §4.1): a real HV stick drops volts even at nA (3.8 V at 1 nA, 10 V at 1 µA), and its capacitance and leakage take z below 1 at small amplitudes. The free pump decays below about 17 V on nodes 1 and 4 (typical leakage), 64 V (the datasheet's maximum) or 260 V (hot), so a contact potential or a triboelectric charge does not start it.
  - On the record's capacitances it needs a seed of tens of volts, a few hundred hot, and the deck's −1 kV is enough at every leakage level. **As built** (the tube's strays solved, `sim/diodes-real-findings.md` §6) it needs 122 V with typical leakage, 1.0 kV at the datasheet's maximum and 3.65 kV hot, so the deck's −1 kV starts it only just at the maximum and not at all hot. The machine's seed source is not designed (§5.2).

**The clamps and the operating point.**

- z > 1 at every amplitude, so something must stop the growth. The clamps Z1 (node 1) and Z4 (node 4) are avalanche strings, first cut 66 × 200 V (1.5KE200A-class parts, one lot, sorted and trimmed warm: `sim/parts-first-cut-findings.md` §2.4). They break down at the operating peak V_op 13.13 kV, conduct once a cycle near the peak and turn the whole surplus into heat: 1.07 W each, 0.95 mA peak (`sim/core_field_results.json`).
- **V_op** is the 6 mm air gap's breakdown, 19.7 kV, over a 1.5 margin [RH].
- **The rims' field** at 13.1 kV is 44.4 kV/cm, 0.80 of Peek's onset (55.7 kV/cm). Corona starts at 16.5 kV on a smooth surface and 14.0 kV on a handled one, margins of ×1.26 and ×1.07 (`sim/air_stack_sizing_results.json`).
- **The nodes:**
  - nodes 1 and 4 swing −13.2 ↔ −5.7 kV, half a cycle apart (mean −8.6 kV);
  - nodes 2 and 3 swing 0 ↔ −6.1 kV.
  - D1–D4 see reverse peaks of 6.1 / 6.1 / 13.2 / 13.2 kV. First cut (PROPOSED): one 20 kV avalanche stick for D1 / D2 and two in series for D3 / D4, each with a 22 kΩ surge resistor (`sim/parts-first-cut-findings.md` §2.2).
- **The power:** the belt pays 2.14 W, all of it into the clamps (0.65 W as built, `sim/tube-strays-findings.md` §5).
- **Without clamps** the pump runs as a relaxation oscillator: it arcs at the stack, sends 25–47 A surges through the diodes and 38 A through the bearing link, so the clamps stay (`sim/electrostatic-no-clamp-findings.md`).

- **Ca / Cb:** 1.1 C_max, six full-annulus aluminium plates per side with 5 gaps of 6 mm, on the rotor. Adjacent plates differ by 5.7–7.5 kV, which needs 40 mm of creepage on PTFE or 80 mm on G10 (design; 32 / 64 mm by the standard) and 13 mm of clearance. So no mount may bridge two adjacent plates: a first cut rides the node-1 plates on the vane stack and hangs the node-2 plates on PTFE-sleeved tie-rods (`sim/parts-first-cut-findings.md` §4). Not drawn.
- **A vane flashover** at node 4, at its lowest point, lifts ring B from 14.96 to 19.0 kV until the leakage drains it: 2 % past the bench's 1.25× hold-off. Before the DC settles, the polar bead's gel would then run at about 6.3 kV/mm (`sim/parts-first-cut-findings.md` §2.7) [IR]. OPEN for the designer (§5.2).
- **Strays:** 20 pF per node [RH] in the deck. Each pF costs about 0.003 of z (`sim/core-field-findings.md` §4; 0.0027 for nodes 1 and 4 together, `sim/tube-strays-findings.md` §5).
  - **Solved** (`sim/tube-strays-findings.md` §3): 66.2 pF on nodes 1 / 4, of which 40 pF is the rotor rings to the shaft through the G10 sleeve and 15–23 pF the first Ca plate to the stator vanes through the last rotor vane's openings; 24.5 pF on nodes 2 / 3, 11.5 pF of it to the utrons. Across the hub, 0.05 pF or less.
  - **As built** (every capacitance solved, the record's deck): z 1.058 bare and 1.031 with the rings' chains, 0.65 W, the rings at ±13.86 kV (7.06 kV/cm at the null), 95 % in 0.63 s: with ideal diodes it barely self-excites.
  - **As built with real HV sticks** (`sim/diodes-real-findings.md` §6): it does not self-excite. It starts from 122 V (typical leakage), 1.0 kV (the datasheet's maximum) or 3.65 kV (hot), and holds −13.38 / +13.40 kV and 6.82 kV/cm with typical sticks (95 % at 0.575 s from −1 kV), 6.31 kV/cm at the maximum leakage. The clamps' own leakage helps decide it: their hot 5 µA a string is [RH] and sets the 3.65 kV.
  - The parts' first cut's conductive spacer rings on the sleeve would raise node 1's stray to 105 pF and take z with the chains to 1.008; insulating mounts avoid it (§5.2).

## 3.5 The hub

Source: `presets/hub-locked.json` (locked 2026-10-08; the decisions of 2026-10-09).

| layer, inside out | what | status |
|---|---|---|
| the vessel | borosilicate sphere, OD 50.0 mm, wall 1.5 mm (R_in 23.5 mm), ε 4.6, vacuum inside; about 0.8 MPa shell stress | DECIDED |
| the rings | two copper-foil bands on the outer surface (§3.6) | DECIDED / DESIGN |
| the AH | the MnZn rods with their 160-turn coils on the axis, PEEK seats at ±25–30.7 mm (§3.3) | DECIDED (placement) / REGISTER (dimensions) / DESIGN (the winding) / PROPOSED (the ends rounded to 1.5 mm) |
| the retainer | unfilled PEEK, ε 3.2, r ≤ 30 mm, ±72 mm; a 0.5 mm pocket over the glass with grooves for the beads | DECIDED (role; material, 10-09) / PROPOSED (the grooves' full-round tops; the dimensions' first cut) |
| the interface filler | two-part silicone gel, degassed and vacuum-cast into the pocket, ε 2.9, 0.5 mm | DECIDED (10-09) |
| the shaft coupler | G10 outside the PEEK, r 30–33 mm, ε 4.7 | DECIDED (role; G10, 10-09) / PROPOSED (a straight tube over the retainer and both flanges) |
| the shaft | split, flanges r 32 × 8 mm at ±72–80 mm; non-magnetic | DECIDED (layout) / REGISTER (d 30) / OPEN (with the bearings) |
| the pumps | each shaft half carries its side's electrostatic stack and utron group, symmetric top and bottom | DECIDED |

- **Why PEEK and gel:**
  - the glass–retainer interface is what holds the DC between the rings, and a dry machined fit leaves air gaps; the gel fills them, beds the beads and takes up the PEEK–glass expansion mismatch;
  - PEEK is non-magnetic and non-conducting, machines well and barely moves the field (−1.5 % to +0.7 % from alumina to PTFE);
  - it is the ten-times-less-conductive PEEK, against the glass and gel, that sets the drift (§3.7) (`sim/hub-rings-build-findings.md` §1).
- **Once the DC has settled** (minutes to hours; `sim/hub-beads-settled-findings.md`), conduction shares it by the conductivities [OC]:
  - the beads relax (0.69 / 1.88 kV/mm in the gel), and the PEEK, ten times less conductive than the gel, takes the DC: 2.3 kV/mm across the AH seat on average;
  - the drawn grooves' square top corners then hold 7 kV/mm in the PEEK and the AH end's square corner 7.9. Proposed: the grooves' tops full-round with 1.0 mm of gel (4.1 kV/mm) and the AH ends rounded to 1.5 mm at REF (4.5 kV/mm), about a fifth of PEEK's short-term strength. The AH seat stays 5.7 mm;
  - **the equatorial beads set a condition:** their gel holds 5 kV/mm only while the glass conducts at most 4.1 times the gel (4.4 with the full-round grooves). With the glass at 40 °C (5.4×) they reach 6.15 kV/mm;
  - the bead–glass contact wedge stays finite, and with the gel in place the voltage across it is at most half of air's Paschen breakdown. A void in the gel's place is not safe once settled: a sealed void reaching 0.5 mm from an equatorial contact is at 0.99 of it in service, 1.46 at 1 mm (`sim/hub-joints-findings.md` §4). The gel fills those wedges void-free.
- **The hub's temperature** (`sim/hub-thermal-findings.md`): the AH coils dissipate 1.22 W each. In a 25 °C room, with the air between the vane stacks 5 K above it [RH], the coils run at 43 °C and the glass at 32–37 °C, 32.5 °C between the rings.
- **The vessel's resistivity,** 1e13 Ω·m at 25 °C [IR], falls with 0.90 eV (the record's two points). Between the rings in service it is 4.2e12 Ω·m, 2.4 times the gel's conductivity: the beads hold, up to a room of about 30 °C if the gel's conductivity does not rise with temperature. The bench measures both (phase 3).
- **The coupler and the leads** (first cut, PROPOSED; `docs/rings-design.md` §2):
  - a straight G10 tube, Ø60 / Ø66 mm over |z| ≤ 80 mm, holding the PEEK retainer and both shaft flanges (turned to Ø60), each bonded and pinned;
  - each ring's lead leaves its equatorial bead along the bead's normal, through the PEEK and the coupler, and runs over the coupler to its own end;
  - the air outside sees at most 0.55 kV/mm.

## 3.6 The rings and their symmetric supply

**The rings** (`docs/rings-design.md`; drawing DCCREG-HUB-201, Figure 6).

- **The bands.** Two bands of annealed Cu-ETP foil, 0.10 mm, lie on the vessel's outer surface from 26.25° to 55.71° from the axis. Ring B is the upper one, ring A its mirror.
  - Each band is 12.85 mm along the glass and 13.1 cm², and the equatorial gap is 29.9 mm along the glass.
  - Each band is cut as 12 gores, 5.79 → 10.81 mm wide, overlapped 1 mm and soldered, and burnished on. There is no adhesive film: the gel bonds it.
  - **The laps** (first cut, PROPOSED; `sim/hub-joints-findings.md` §2): each upper gore's free edge is a 0.1 mm step. As cut it lifts the gel's 1.20 kV/mm at switch-on to 3.5–4.4 kV/mm; under a solder fillet 0.2 mm wide, to 1.9. So the edges are deburred, each lap's free edge filleted and each lap burnished into the gel film, with no void under it.

- **The edges are beaded.** A bare foil edge concentrates the field, so each edge carries a soldered copper wire ring:
  - Ø3 mm at the polar edge, 73.6 mm along its centre line;
  - Ø2 mm at the equatorial edge, 135.0 mm;
  - **the closing joints** (first cut, PROPOSED; §3 there): a joint left proud raises the bead's field by about 1 + 3.7 δ / w. The equatorial rings take one dressed to the round within 0.03 mm over 1 mm. The polar rings, 0.3–0.4 % under their rating, take none: they are best seamless (turned), or the AH ends rounded to 1.5 mm, which gives them 2.3–2.4 %;
  - each bedded in a groove of the PEEK, 3.5 / 2.5 mm deep (proposed: full-round tops with 1.0 mm of gel, 4 / 3 mm deep).
- **Why these angles.**
  - The polar bead faces the AH coil's end across 5.4 mm of PEEK. A bigger bead comes closer to it, so the polar edge does not ease with size, and a ring at 15 kV needs its polar edge at 26° or more.
  - The gap across the equator holds the DC at 1 kV/mm along the glass.
  - Both beads hold 5 kV/mm in the gel [RH]: 4.98 / 4.46 kV/mm as drawn (4.85 / 4.36 in the build's box, which had gel throughout). The polar bead sits 0.4 % under its rating.
  - The rings' separation is not the limit; the beads are (`sim/hub-rings-build-findings.md` §2, §4).
- **Later:** a fired-on coating needs the same beads, or a resistive grading toward the pole.

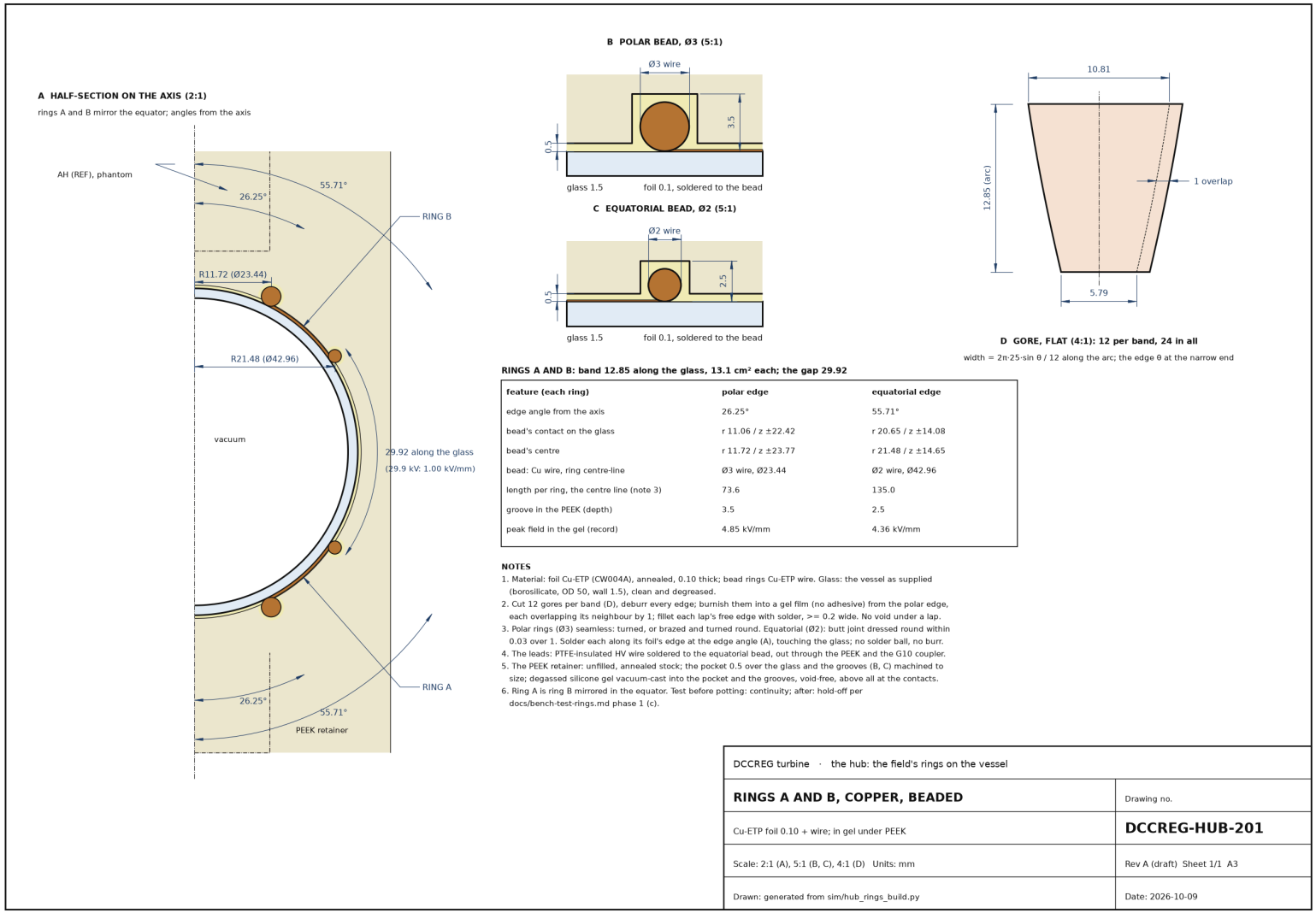
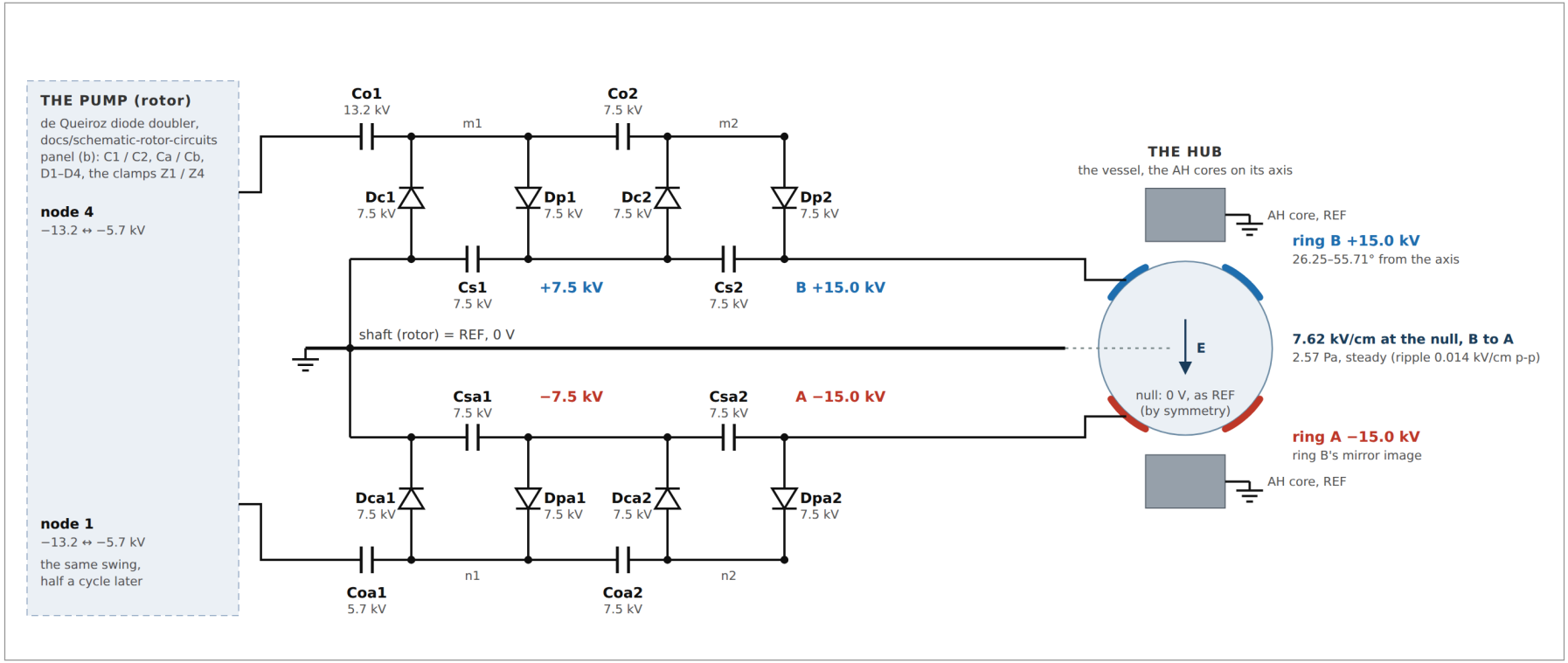



Figure 6. DCCREG-HUB-201, the rings' manufacturing drawing: the half-section at 2:1, the beads in their grooves at 5:1, one gore flat at 4:1, the feature table and the build notes. Source: docs/drawings/DCCREG-HUB-201.pdf (docs/make_rings_drawing.py).

### DCCREG turbine: the rings' DC supply of record, the symmetric pair (each ring on its own chain from the shaft)

Ring B on 2 positive Cockcroft-Walton stages driven by the pump's node 4, ring A on the mirror image driven by node 1; both chains stand on the shaft (REF), which the AH and the null share. 100 pF storage, 100 GΩ per ring.




| | |
|---|---|
| **chains** | 2 stages each; ring B's from node 4 (Co, Dc, Dp, Cs), ring A's from node 1 (Coa, Dca, Dpa, Csa), the diodes reversed |
| **levels** | each stage adds node 4's / node 1's swing: 7.5 kV a stage at 100 GΩ per ring; 29.9 kV across the rings |
| **diodes** | 8 HV stacks, reverse ≤ 7.5 kV each (20 kV stacks, derated 1.5×) |
| **capacitors** | 8 × 100 pF / 30 kV; each holds about one stage's 7.5 kV DC, but Co1 13.2 and Coa1 5.7 kV: nodes 4 and 1 sit at −8.6 kV mean |
| **ripple** | ring A 34 V, ring B 29 V p-p at 120 Hz; ring A tops up as node 1 bottoms, ring B as node 4 peaks, 0.124 of a cycle later: 57 V p-p across the gap, 0.2 % of the field; no swing |
| **pump** | belt 2.14 W into the clamps; leakage 4.5 mW; start-up gain z 1.191 |
| **rings** | Cu foil bands, beads Ø3 / Ø2 mm (polar / equatorial) in gel; strays 5.1 pF each to REF, 1.1 pF between |

Numbers: sim/hub_rings_build_results.json (record, record_supply), sim/hub_drift_results.json (swing), sim/core_field.py (the deck) · findings: sim/hub-rings-build-findings.md · generator: docs/make_rings_supply_schematic.py

Figure 7. The rings' DC supply of record: ring B on two positive Cockcroft-Walton stages from node 4, ring A on the mirror image from node 1; both chains stand on the shaft; each capacitor's DC and each diode's reverse peak. Source: docs/schematic-rings-supply.svg (docs/make_rings_supply_schematic.py).

**The supply: two mirrored Cockcroft-Walton chains** [OC] (Figure 7; `sim/core_field.py` dc, `n_cw = n_cw_a = 2, a_ref = "shaft"`).

- **Ring B's chain:**
  - node 4 → Co1 → m1 → Co2 → m2;
  - clamp diodes Dc1 (shaft → m1) and Dc2 (b1 → m2), peak diodes Dp1 (m1 → b1) and Dp2 (m2 → ring B);
  - smoothing capacitors Cs1 (shaft–b1) and Cs2 (b1–ring B).
  - Each stage lifts the smoothing column by node 4's swing, 7.5 kV.
- **Ring A's chain** is the same ladder on node 1 with every diode reversed (Coa, Dca, Dpa, Csa). It lowers its column by 7.5 kV a stage.
- **How it pumps against the core centre.**
  - Both chains stand on the shaft (0 V). The hub is mirror-symmetric and ring A is ring B's negative, so the null — and the whole equatorial plane — sits at the shaft's potential with nothing connected to it [OC].
  - Each ring stands 15 kV from it, as far as its polar bead allows.
- **The parts and levels:**
  - 8 HV diode stacks, all at 7.5 kV reverse;
  - 8 capacitors of 100 pF / 30 kV, each holding a stage's 7.5 kV DC, except Co1 at 13.2 kV and Coa1 at 5.7 kV (nodes 4 and 1 sit at −8.6 kV mean);
  - the levels: b1 +7.49 kV, ring B +14.96 kV, a1 −7.49 kV, ring A −14.96 kV; about 27 mJ stored;
  - **with real HV sticks** (`sim/diodes-real-findings.md` §4.3) the rings stand at −14.65 / +14.67 kV: the sticks' 0.2–0.4 pF of junction capacitance across each chain diode takes 0.57 kV off the gap. At the datasheet's maximum leakage (2 µA) the gap is 27.3 kV, hot (5 µA) 24.3 kV, so the chain diodes are selected for ≤ 25 nA at 7.5 kV and kept cool;
  - no Dk and no C_A (`sim/hub_rings_build_results.json` record_supply).
- **Why 100 pF:** it keeps the pump's start-up gain at 1.191 (1.03 at 1 nF), and still smooths the ripple to tens of volts. As built, with the tube's strays solved, the gain with the chains is 1.031 (`sim/tube-strays-findings.md` §5).
- **The strays:** 5.1 pF from each ring to REF and 1.1 pF between the rings, behind diodes. The whole machine solved gives 5.87 / 1.50 pF, and the deck ran 3 / 2 pF; none of it moves the supply (`sim/tube-strays-findings.md` §3).

## 3.7 The field at the null, in space and in time

**How the field is computed** [IR] (`sim/core_rings.py`, `sim/hub_locked.py`, `sim/hub_rings_build.py`).

- An axisymmetric finite-volume solve of div(ε grad V) = 0 on the half hub, with 0.25 mm cells.
- REF is the AH envelope, the flanges, the shaft and the box.
- Each geometry is solved in two modes:
  - anti: the rings at ±1, which gives the field at the null per volt across;
  - sym: both at +1, which gives the strays.
- Any supply is their superposition, exact for this linear problem [OC]: E_null = k (V_B − V_A), with k = 0.2546 (kV/cm)/kV as built (0.2373 for the bands alone).
- The beads come from local solves with 0.025 mm cells, bounded by the hub's solution, and as drawn they also sit in the hub's maps (`sim/hub_beads_settled.py`).

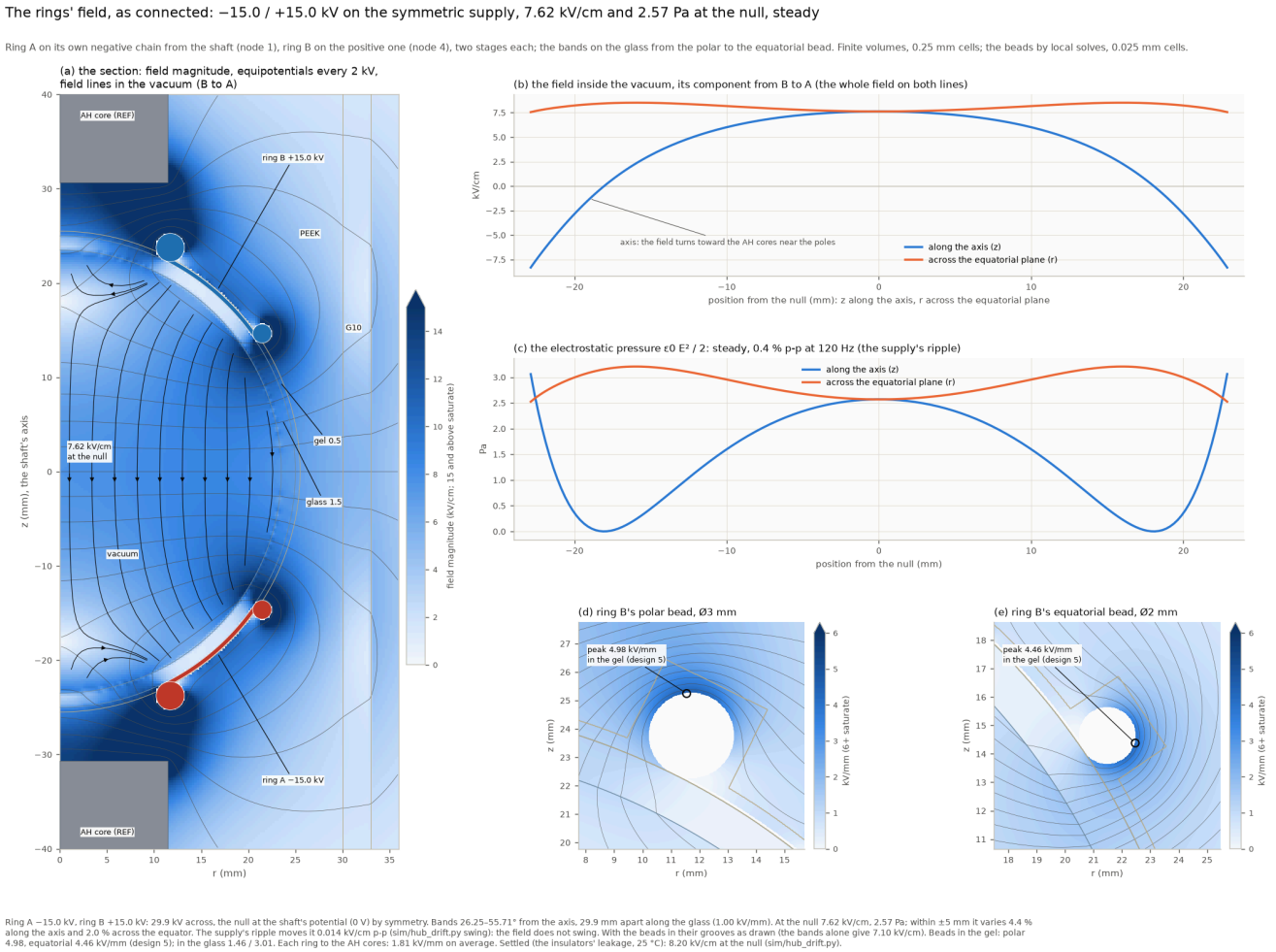



Figure 8. The rings' field: |E| in the section with equipotentials and field lines from B to A; the field and the pressure along the axis and across the equator; ring B's beads from local solves. Source: docs/figures/hub-rings-field.png (docs/make_rings_field_figure.py).

**In space:**

- **At the null:** 7.62 kV/cm along −z (from B to A) and a pressure ε0E²/2 of 2.57 Pa, at switch-on. The beads add 7 % to the bands' 7.10: they are at the ring's potential and reach past its edges toward the gap.
- **Within ±5 mm** it varies 4.4 % along the axis and 2.0 % across the equator.
- **Further out:** across the equator it rises to 8.5 kV/cm at r 16 mm. Along the axis it falls to zero at |z| ≈ 18 mm and then turns toward the AH cores, because each band sits between the null and an AH core, both at 0 V.
- **In the insulation,** at switch-on:
  - 1.00 kV/mm along the glass between the rings, on average;
  - 4.98 / 4.46 kV/mm in the gel at the beads, as drawn;
  - 1.46 / 3.01 kV/mm in the glass under them;
  - each ring averages 1.81 kV/mm to the AH cores.
- **The retainer's ε** moves the field at the null by less than 2 %; the AH cores take about 13 % of it.

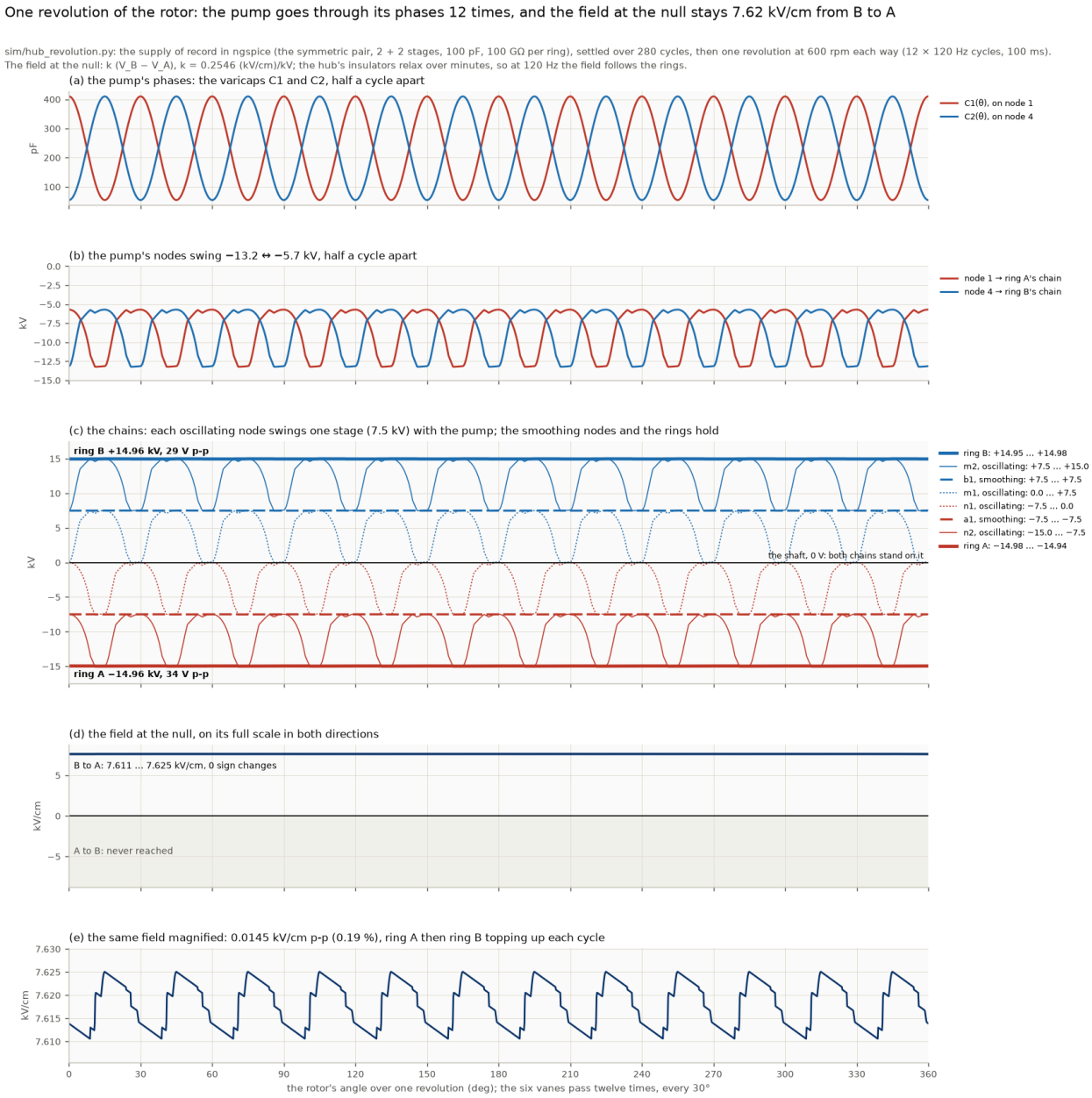



Figure 9. One revolution of the rotor: the pump goes through its phases twelve times (C1 / C2, nodes 1 / 4, the chains' oscillating nodes), while the rings hold and the field at the null stays 7.611–7.625 kV/cm from B to A. Source: docs/figures/hub-rings-revolution.png (sim/hub_revolution.py).

**In time** (the bench test's predictions are Figure 12, §5.3):

- **Start-up** from the pump's seed: 50 % at 0.12 s, 95 % at 0.24 s, 99 % at 0.33 s; as built, with the tube's strays solved, 95 % at 0.63 s on the deck's own start (`sim/tube-strays-findings.md` §5); with real HV sticks as well, 0.575 s from a consistent −1 kV, 1.49 s at their maximum leakage, and hot no start from −1 kV (`sim/diodes-real-findings.md` §6.4).
  - **Corrected 2026-10-09** (`sim/diodes-real-findings.md` §1.2, §4.2): the deck's 29 cycles start from −0.58 / −0.90 kV, not −1 kV, because under `uic` ngspice starts its charge-defined varicaps uncharged. From a consistent −1 kV, 95 % takes 27 cycles (0.225 s), with ideal diodes and typical sticks alike; 0.30 s with hot sticks.

- **The ripple:**
  - ring A carries 34 V p-p and ring B 29 V p-p at 120 Hz;
  - ring A tops up as node 1 bottoms; ring B tops up 1.03 ms later, as node 4 peaks;
  - so the two mostly add: 57 V p-p across the gap, 0.014 kV/cm p-p (0.19 %) at the null;
  - with real HV sticks their capacitance triples it: 193 V p-p, 0.049 kV/cm p-p (0.66 %) (`sim/diodes-real-findings.md` §4.3).
- **No swing.** Over a whole revolution (12 pump cycles in 0.1 s) the field never changes sign.
  - Each chain's diodes pass charge one way only, and the storage capacitors hold the rings between cycles.
  - The hub turns with the rotor and the rings are symmetric about the shaft, so the rotation changes nothing at the null either.
  - A field swinging from A to B would need AC on the rings, through coupling capacitors. That gives about ±1.8 kV/cm, with the pressure peaking near 0.14 Pa [IR estimate, not simulated with the rings].
- **The drift.** With the rings held at their DC, the insulators' leakage moves the potential from the electrostatic toward the conduction-settled state [OC]. The conductivities are datasheet-class [IR].
  - PEEK with gel at 25 °C: 7.62 → 7.95 (10 min) → 8.18 (1 h) → 8.20 kV/cm (6 h), +7.6 %, half-way at 7.6 min.
  - PEI gives +10.5 %, G10 +2.0 % (its σ/ε matches the glass's), and the glass at 40 °C +8.5 % in about a third of the time.
  - Settled, the beads relax and the PEEK takes the DC (§3.5).
- **The leakage:** the ledger of paths gives 158 GΩ per ring, against a 100 GΩ estimate. The chains' diodes and capacitors dominate; the rings' own insulation is about 200 TΩ (`sim/hub-rings-build-findings.md` §3).

## 3.8 The cost

Source: `docs/cost/README.md`. Every price is a placeholder for a one-off prototype, in EUR [RH].

**The sheet** (`docs/cost/dccreg-air-build-cost-sheet.xlsx`) costs all 2700 vane-matrix stacks with live formulas.

**The placeholder targets:**

- 6.5 kV/cm at the null, which needs 27.4 kV across the rings on the symmetric supply (on the bands' 0.2373: the rings as built give 7 % more, so the picks are conservative);
- at least 2 W, with z ≥ 1.3;
- at most 6 + 6 vanes;
- corona-safe rims;
- 300 steady A-turns.

81 designs meet them.

| build | design | power | total |
|---|---|---|---|
| cheapest that qualifies | 6 mm gaps, 2.5 mm vanes, r 175 mm, 22°, 4 + 4 | 2.01 W | 6,060 |
| the stack of record (the designer kept it, 10-09) | 6 mm gaps, 3 mm vanes, r 150 mm, 22°, 6 + 6 | 2.14 W | 6,477 |
| lowest cost per watt | 8 mm gaps, 4 mm vanes, r 300 mm, 26°, 6 + 6 | 16.4 W | 8,368 (510 per W) |

**The fixed parts** are 3,754 before the 15 % contingency, about 70 % of the cheapest build (the BOM corrected at the settlement: one 22 mF part per coil, six spiders, the utron's SiFe as built):

- magnetic pump 1,552;
- mechanics 1,282, including the gear 250, the drive 250 and the frame 300;
- hub 626;
- electrostatic 180;
- AH 114.

**The drive's first cut** costs 1,066 against the sheet's 800 for the gear, the drive and the frame (`docs/drive-gear-belt.md`). The sheet keeps its placeholders until the designer accepts it.

**The stacks' cost** is per-part work, not metal (metal is about 8 %). Aluminium is the choice.

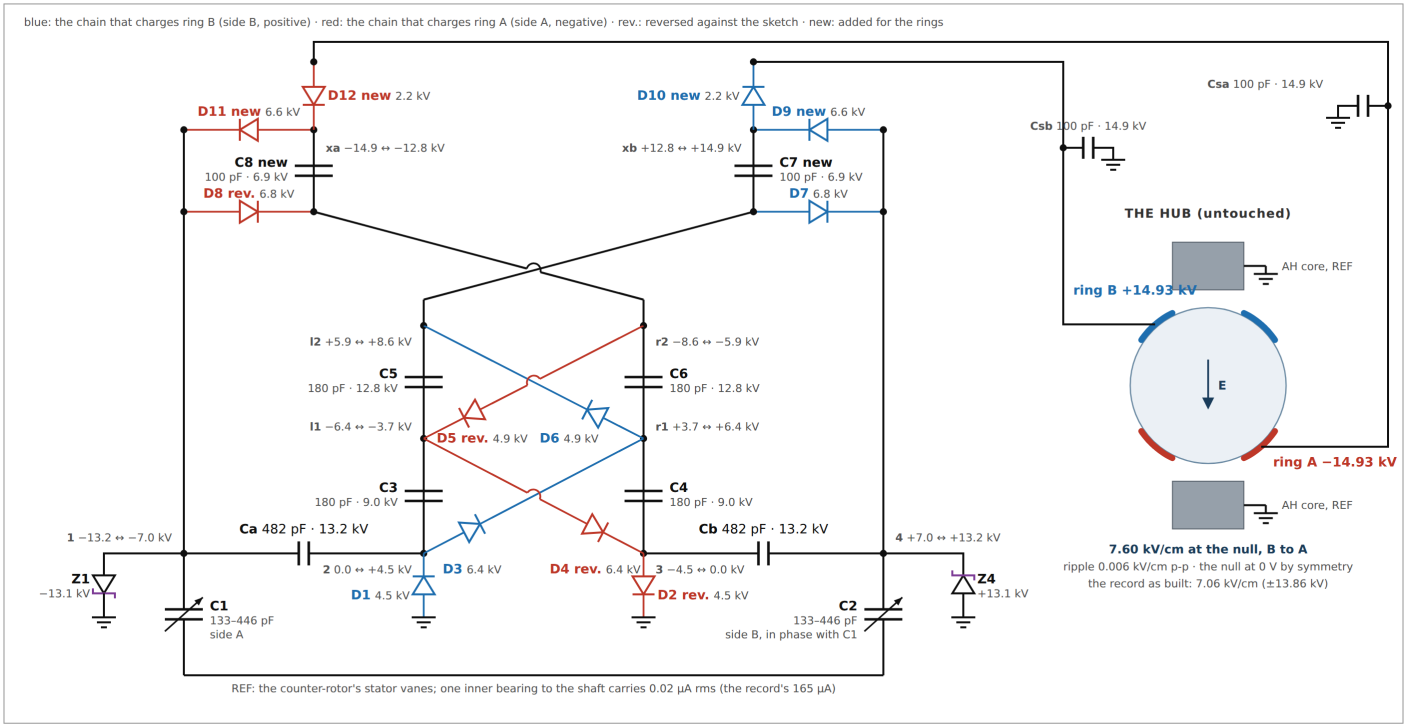
**The symmetric supply** costs 50 more than the asymmetric one: three more diodes and three more capacitors.

**Not costed:** re-sizing the AH or the pumps, the shaft for larger radii, design, test equipment, the vacuum system and tooling.

## 3.9 The redesign: the multiplier stacked into the doubler (2026-10-10, PROPOSED)

**DCCREG turbine: the multiplier stacked into the doubler, run bipolar, feeding the rings — PROPOSED (2026-10-10)**

The designer's ladder C3–C6 / D5–D8 as sketched; chain 2 reversed (D2, D4, D5, D8): side A negative, side B positive; C1 / C2 in phase; one stage past each varicap node to its ring. As built, ideal diodes.




| | |
|---|---|
| **circuit** | the designer's ladder (C3–C6 180 pF, D5–D8), bipolar: chain 2 reversed; C1 / C2 in phase (in antiphase it does not pump); each chain one stage past its varicap node to its ring |
| **levels** | nodes 1 / 4 at the clamps' −13.2 / +13.2 kV (V_op of record); the rings −14.93 / +14.93 kV, 7.60 kV/cm at the null (the record as built ±13.86 kV, 7.06 kV/cm) |
| **diodes** | 12 HV sticks, one a position, reverse ≤ 6.8 kV (the record: 14, its D3 / D4 13.2 kV on two each) |
| **capacitors** | Ca / Cb now hold 13.2 kV (the record's 6.9); C5 / C6 12.8, C3 / C4 9.0, C7 / C8 6.9, Csa / Csb 14.9 kV |
| **pump** | z 1.0142 as built (the record with its chains 1.0322); the clamps 630 mW; 95 % in 1.45 s; the counter-rotor can float (Q1 = −Q2) |
| **real sticks** | seeds 51 V / 947 V / 4870 V (typical / maximum / hot; the record 122 V / 1.0 kV / 3.65 kV); the rings ±14.67 / ±14.40 / ±13.95 kV, 7.47 / 7.33 / 7.10 kV/cm |
| **open** | the new nodes' strays (24.5 pF each assumed) set the rings: with none they reach ±16.3 kV; trim the clamp strings on the bench before the hub sees it |

Numbers: sim/integrated_multiplier_results.json (pick, as_built, real) · findings: sim/integrated-multiplier-findings.md · generator: docs/make_integrated_multiplier_schematic.py

Figure 10. The multiplier stacked into the doubler, as proposed: the designer's ladder (C3–C6, D5–D8) as sketched, run bipolar (chain 2 reversed) with C1 / C2 in phase, each chain continued one stage past its varicap node to its ring; every node's swing, each diode's reverse peak and each capacitor's DC as built. Source: docs/schematic-integrated-multiplier.svg (docs/make_integrated_multiplier_schematic.py).

**The brief** (the designer, 2026-10-10; §2.3). The record's multipliers are "a separate circuit tapping on node 1 and 4. It's no good this way." They are to stack into the doubler; the turbine's stack will change, and the magnetic pump, the sphere, its AH and the field interface stay as they are for now.

**The designer's circuit** (the sketch, read back and confirmed) [OC]:

- the record's doubler with each cross diode replaced by a three-diode chain through a two-stage capacitor ladder: C3 (2–l1) and C5 (l1–l2) stand on node 2, C4 (3–r1) and C6 (r1–r2) on node 3;
- the chains 0 → D1 → 2 → D3 → r1 → D6 → l2 → D7 → 4 and 0 → D2 → 3 → D4 → l1 → D5 → r2 → D8 → 1, positive as drawn.

**As drawn** (`sim/integrated-multiplier-findings.md` §1) [OC]:

- it pumps: z falls from 1.31 to 1.00 as C3–C6 grow from 1 pF to 10 nF (2-D), and with small ladder capacitors it is the record's doubler;
- it splits the record's 13.2 kV cross diodes into 4–7 kV steps;
- but it cannot feed the rings: every node sits on one side of the shaft, and none rises above nodes 1 / 4, which the clamps hold at V_op.

**How it can work** (PROPOSED; §2–§4 there):

- **bipolar:** chain 2 reversed (D2, D4, D5, D8). Side A runs negative, side B positive, and the null stays at the shaft's potential;
- **C1 / C2 in phase:** C2's rotor sectors aligned with C1's, not half a pitch apart.
  - With opposite charges the nodes swing against each other only so; in antiphase it does not pump (z 1.000 as built).
  - Every mirror-symmetric four-diode doubler peaks at 1.098 in phase (2-D), against the record's unipolar 1.31: a bipolar pump's clamp diodes give back part of each transfer;
- **one stage past each varicap node to its ring:**
  - ring B on C7 (from l2), D9 (4 → xb) and D10 (xb → ring B); ring A on the mirror image, C8 (from r2), D11, D12;
  - each ring on 100 pF to the shaft;
- **C3–C6 at 180 pF:** the largest whose rings stay at the record's ±14.96 kV as built (§4 there).

**As built** (the solved strays, the record's V_op; ideal diodes unless stated):

| | the record's supply | the proposal |
|---|---:|---:|
| gain z at start-up | 1.0322 | 1.0142 |
| the rings; the field at the null | ±13.86 kV; 7.06 kV/cm | ±14.93 kV; 7.60 kV/cm |
| typical sticks: z from 1 kV, seed, rings | 1.0244, 122 V, ±13.39 kV | 1.0129, 51 V, ±14.67 kV |
| maximum leakage | 1.0038, 1.0 kV, ±12.38 kV | 1.0039, 947 V, ±14.40 kV |
| hot | 0.9672, 3.65 kV, ±10.80 kV | 0.9870, 4.87 kV, ±13.95 kV |
| HV sticks | 14 (D3 / D4 two each, 13.2 kV) | 12, one each, ≤ 6.8 kV |
| Ca / Cb's DC | 6.9 kV | 13.2 kV |
| the reference link | 0.165 mA rms | 0.02 µA rms |
| the varicaps' 120 Hz torque | 63 mN·m p-p | 191 mN·m p-p |
| 95 % of the rings, from 1 kV | 0.53 s | 1.45 s |

**What it gives and costs:**

- the rings at the record's level as built, and higher than the record's with real HV sticks at every leakage. The gain from 1 kV is lower at typical leakage, level at the maximum and ahead hot;
- no current through the bearing link: Q1 = −Q2, so the counter-rotor can float (on 100 pF it pumps the same);
- Ca / Cb at 13.2 kV DC, so their mounts' creepage roughly doubles (§3.4);

- the rings' level rests on the new nodes' strays, 24.5 pF assumed [RH]: with none the rings reach ±16.3 kV. Set the clamp strings on the bench (each 200 V part moves V_op 1.5 %) before the hub sees them;
- in phase, the stacks' pulls add: the varicaps' torque ripple triples;
- not run: a vane flashover with the ring stages, the torque's share at the gear.

**The mechanical layout** (2026-10-11, PROPOSED; `sim/multiplier-layout-findings.md`). The designer took the circuit "for now" and asked for its capacitors laid out top and bottom of the stack, the diodes placed, a mechanical layout.

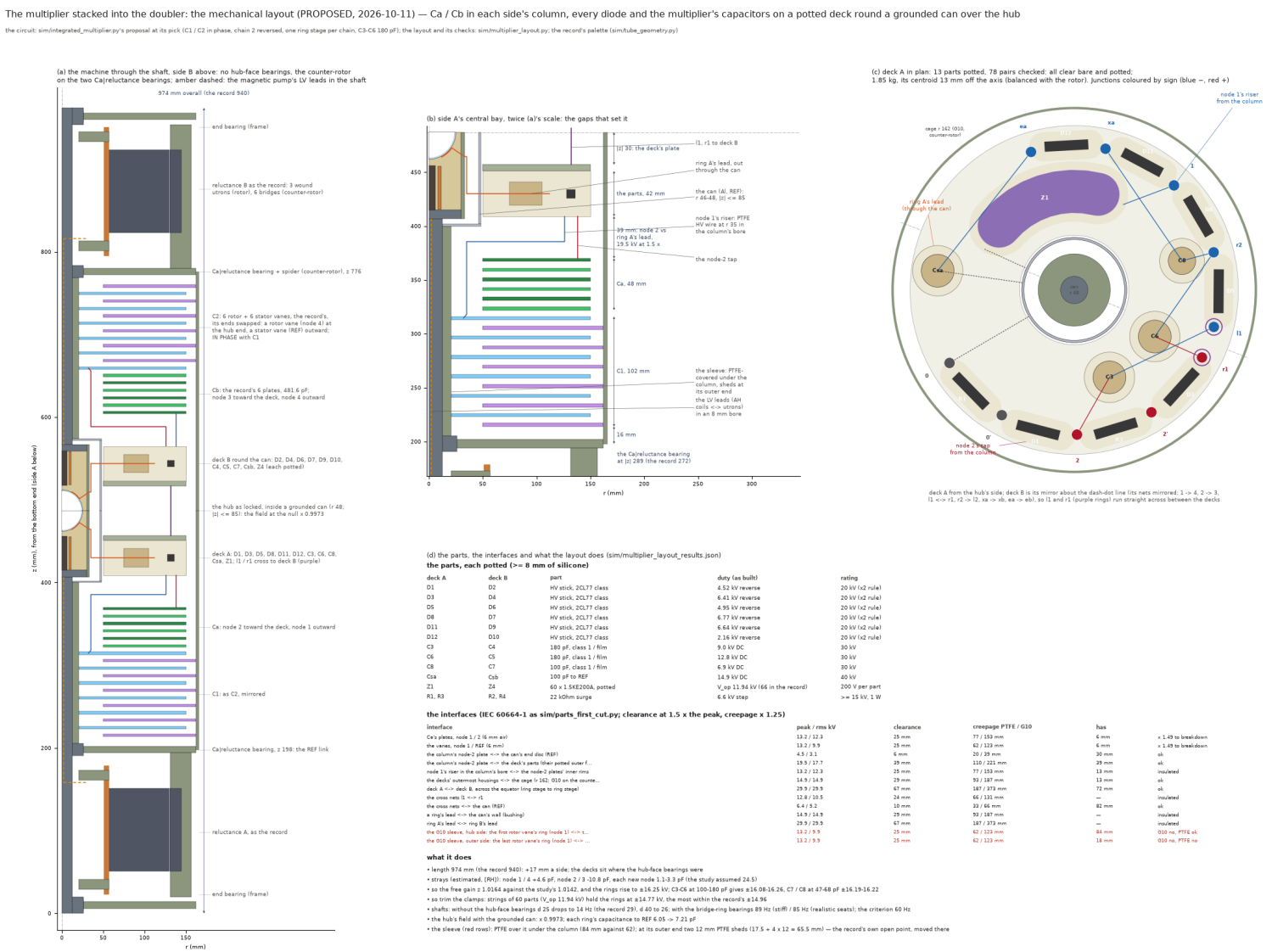



Figure 11. The multiplier laid out in the machine (PROPOSED): (a) the half-section, side B above, without the hub-face bearings; (b) side A's central bay with the gaps that set it; (c) deck A in plan, the sticks along the outer lane, the capacitors and the clamp string on the inner; (d) the parts, the interfaces and what the layout does. Source: docs/figures/multiplier-layout.png (docs/make_multiplier_layout.py; sim/multiplier_layout.py).

- **The stack, per side from the hub outward** (|z| from the hub's centre; side B the mirror) [IR]:

| |z| (mm) | what |
|---|---|
| 0–80 | the hub as locked, in a grounded aluminium can on the rotor (r 46–48, \|z\| ≤ 85) |
| 30–78 | **the deck**: a G10 ring plate (r 50–150) round the can carrying the side's six diodes, C3–C8 and Csa / Csb, the clamp string and two surge resistors, each potted |
| 117–165 | **the column**: Ca (Cb), the record's six plates, node 2 (3) toward the deck |
| 171–273 | **the vane stack**: the record's 6 + 6, its ends swapped (a rotor vane at its hub end) |
| 289 | the Ca\|reluctance bearing (the record's 272), then the reluctance section as the record |

- **Why there are no hub-face bearings** [OC]:
  - a counter-rotor spider leaves no path for a rotor wire except the shaft;
  - nodes 1 / 4 in a shaft bore would take 3.0 pF each, 0.008 of z;
  - the record's own wiring (D3 / D4, the chains, the rings' leads) has the same unsolved crossing.
- **The diodes' placement:**
  - deck A holds D1, D3, D5, D8, D11, D12 with C3, C6, C8 and Csa; deck B holds the mirror set;
  - **only l1 and r1 cross** between the halves, the fewest the ladder's two odd cycles allow, on PTFE HV wire between the decks;
  - C3–C8 are discrete (class 1 HV ceramic or film): as plates they would put a third node in each column.
- **Clearances** [IR]:
  - every one of the 78 part pairs per deck, and every part to the can, clears the record's bare clearance;
  - the potting adds at least 16 mm of silicone;
  - the column stands 39 mm off the deck (node 2 against ring A's lead, 19.5 kV);
  - deck A and deck B are 72 mm apart (29.9 kV needs 67);
  - **the sleeve:** a PTFE cover under the column, and two 12 mm PTFE sheds at its outer end. That end holds only 17.5 mm, against 62 mm of creepage on PTFE. This is the record's own open point, moved there.
- **The hub:** with the can as its REF boundary, the record's hub solve gives the field at the null × 0.9973. The can also screens the hub from the decks' 120 Hz swing.
- **The circuit with the layout's strays** (estimated [RH]):
  - **the strays:** nodes 2 / 3 lose 10.8 pF, nodes 1 / 4 gain 4.6 pF, and the new nodes carry 1–3 pF against the 24.5 assumed;
  - **the gain** at start-up rises to 1.0164 (from 1.0142);
  - **the rings rise to ±16.25 kV,** and neither C3–C6 nor C7 / C8 brings them down;
  - **trim the clamp strings to 60 parts** (V_op 11.94 kV): the rings hold ±14.77 kV, the field 7.50 kV/cm.
- **The shafts** (the record's two-body model, its solids moved; criterion 60 Hz):
  - without the hub-face bearings, d 25 falls from 29 Hz to 14 Hz and the utron gap moves 499 µm per g;
  - d 40 alone reaches only 26 Hz;
  - **with the record's bridge-ring bearings: 89 Hz** (85 on realistic seats) and +1 µm. They become necessary; as placed and drawn (below) they give 87 / 83 Hz.
- **The length:** 974 mm (+34 mm); 1014 mm with the bridge-ring bearings (below).
- **The magnetic pump's LV leads** (the AH coils to the utrons' circuit) cross the Ca|reluctance spiders in an 8 mm bore in each shaft half. The record leaves them unrouted.

**The layout in solids, and its circuit running** (2026-10-11, PROPOSED as the layout; `sim/multiplier-layout-3d-findings.md`, `sim/multiplier-waveforms-findings.md`; sheets 32–33; the viewer

`tools/step-viewer/layout.html`):

- **the solids** (`sim/tube_geometry.py --layout`):
  - 386 of them from 147 prototypes, 1014 mm with the bridge-ring bearings, drawn from the layout study's results, the bearings' study and the record's own parts;
  - every part of both decks, the can, the column, the vanes with C2 in phase, the wiring as routed (the riser, the tap, the cross nets, the rings' leads through the can, the LV cable in its bore), the sleeve's cover and sheds;
  - no clash, no sweep hit; every deck pair matches the study's plan in 3-D (the least housing gap 1.86 mm, as the study);
- **the circuit running** (`sim/multiplier_waveforms.py`; the layout study's deck at its picks, ideal diodes):
  - the rings ±14.77 kV, 7.50 kV/cm at the null with the can, 11.8 V of ripple; 95 % at cycle 146 (1.2 s) from a 1 kV seed;
  - each chain passes 208.5 nC a cycle (25.0 µA): the clamps take 207.2 nC, the rings' stages 1.27 nC (their leakage);
  - through a cycle (0° the vanes aligned): D5 / D6 at 10.5–22.8°, D1 / D2 at 22.7–26.3°, the clamps at 25.2–30.2°, D3 / D4 and D7 / D8 at 40–60°, the ring stages in two short pulses; reverse peaks 3.4–6.1 kV;
  - read by Kirchhoff's law along each chain: an ideal diode's own current at a solver step is not the charge the step moved [OC].

**The bridge-ring bearings** (2026-10-11, PROPOSED; `sim/bridge-ring-bearings-findings.md`; sheets 34–35; the viewer's "Ring bearing A"):

- **the part: SKF 6205-2Z.**
  - It is a 25 / 52 / 15 mm SKF Explorer bearing, shielded both sides, with normal clearance: C 14.8 kN, C0 7.8 kN, a limiting speed of 14 000 r/min, running accuracy P5.
  - It is the record's 6205 class at the record's ratings, with the shields the record asks for.
  - Alternates: FAG 6205-C-2Z and NSK 6205ZZ.
  - The catalogue data come from a web search [IR]; re-read them before ordering.
- **the place:**
  - each in the record's end-bearing slot, the end bearing (frame) one 20 mm slot out, so the layout is 1014 mm;
  - the seat: a stainless cartridge in an 8 mm G10 spider centred in its slot, with a 6 mm G10 ring to the bridge ring's outer face. That keeps 9 mm from the utrons' end turns, where an end disc on the face would leave 3.
- **the shafts at d 25:**
  - 87 Hz on stiff seats, 83 Hz on G10 spiders (the drawn solids agree to 1.2 %); the utron gap +1–2 µm under 1 g;
  - the first mode is now the shaft's central span between the Ca|reluctance bearings, so the diameter moves it (d 30: 96–105 Hz) and the seats barely do;
  - d 40 on G10 spiders puts f3 3.4 % under 240 Hz.
- **the duty:**
  - the bottom one locates the counter-rotor: 331 N at 1200 rpm relative, L10 217 000 h;
  - the top one floats on the record's 100 N preload (SKF's rule for spring preloads gives 125–250 N at d 25);
  - the pair adds 2.1 W of friction (6.1 W at the record's high end) and 0.06 W of windage.
- **the drive's first cut** makes its gear-end bearing this pair's bottom one. But it seats it under an end disc on the ring's face, on the record's 940 mm stack-up; to be re-cut on the layout (§6, item 71).

# 4. Fact sheet: the numbers of record, with their sources

### Operating point and machine

| item | value | source |
|---|---|---|
| speed | 600 rpm each way, 1200 rpm relative | `sim/pole-design-findings.md` §6 |
| pump frequency | 120 Hz (6 sectors and 6 bridges per relative revolution) | `sim/core_field.py`; `sim/pole_design_variants_op.json` |
| tube length | 940 mm, the record in solids (153 solids, no clash) | `sim/tube-shaft-findings.md` §0; `sim/tube_geometry_record_results.json` |
| masses | rotor 30.4 kg, counter-rotor 33.1 kg from the solids, without La / Lb and the small parts; the other studies' own counts, with them, give 32.9–35.3 and 33.4–34.9 kg | `sim/shaft_bearings_record_results.json` (corrected 10-11, §6 item 70); `sim/rotor-mechanics-findings.md` §6; `docs/drive-gear-belt.md` §3 |
| bearings | 6 × 6205-class (25 / 52 / 15 mm) on 8 mm G10 spiders; 4 inner, 2 end; the counter-rotor floats on the 4 inner | `sim/stack_sizing.py`; `sim/tube-shaft-findings.md` §0 |
| the shaft's modes | as laid out, d 25: f1 27–29 Hz, 123 µm gap change under 1 g lateral; d 40: 65 Hz, 25 µm; the eighth bearing pair: 131–165 Hz, 2–3 µm | `sim/tube-shaft-findings.md` §0; `sim/rotor-mechanics-findings.md` §4 |
| shaft | d 25 (tube model) or d 30 (hub register), OPEN; non-magnetic, $\mu_r \leq 1.05$ | `sim/stack_sizing.py`; `presets/hub-locked.json`; `sim/ah-null-findings.md` §5 |
| the gap budget | total runout per inner bearing ≤ 0.027 mm as laid out (P5, axial spring preload), 0.05 mm with the eighth pair | `sim/rotor-mechanics-findings.md` §5 |
| rotating loads | no banding (factors ≥ 19 at 750 rpm); balance G2.5 per body; windage about 12 W, bearings 5 W | `sim/rotor-mechanics-findings.md` |
| reference link | one inner bearing; 0.41 mA rms AC, no DC | `sim/core_field_results.json` |
| the drive (PROPOSED) | bevel reverser at side A, POM-C pinions; HTD 5M 4 : 1, 200 W motor; 0.48 / 0.52 N·m on the bodies, 1.00 into the frame; run-up 32 s | `docs/drive-gear-belt.md` |
| the layout's bridge-ring bearings (PROPOSED) | SKF 6205-2Z at each bridge ring's outer end; the layout 1014 mm; d 25: 83–87 Hz, the gap +1–2 µm under 1 g | `sim/bridge-ring-bearings-findings.md` |

### Magnetic pump (per side unless stated; `sim/pole-design-findings.md` §7–§8, `sim/pole_design_variants_op.json`)

| item | value |
|---|---|
| utrons | 3 at 0 / 120 / 240° on two 12 mm G10 carrier discs; r 57–130 mm; tips 14 mm; slot 30 × 30 mm; stack 100 mm of M235-35A 0.35 mm |
| coil | 200 turns of 1.89 mm² (Ø1.55 mm), 50 % fill, mean turn 319 mm; R 0.58 Ω; L 81.0 / 9.4 mH (κ 8.6, the 2-D screen); in 3-D 83.3 / 12.4 mH (κ 6.70; 5.96–6.44 with the coils coupled); τ 0.140 s |
| neck (clamp) | 80 % NiFe, 3.0 × 100 mm; Ψs 0.134 Wb-turns per group; as built (a stacking factor of 0.90 [RH]) 0.1268, with the aligned L 10 % lower at the laps (`sim/neck-nonlinear-findings.md`) |
| bridges | 6 M235-35A sectors, r 130.5–144.5 mm, 25.5° face, 0.65 kg each, in a G10 ring r 131.5–156.5 mm; side B offset 30° |
| gap | 0.500 mm aligned, 9.74 mm unaligned; 0 clashes in 457 pairs |
| group | 3 utrons in series: 0.243 H; 0.87–2.81 A per branch; 101 V peak |
| La / Lb | 0.146 H DC chokes, 0.29 Ω in the deck, 0.87–1.15 A, ≤ 110 V; first cut EI-84 × 35 mm, 150 t of Ø1.40 mm, 0.276 Ω, 1.66 kg (PROPOSED) |
| D1*–D4* | 0.54 V at 1 A in the deck (a 100 V Schottky's drop; the 200 V class drops 0.71 V); 1.9 / 2.3 A peak; reverse 113 / 104 / 58 / 60 V with the start-up; first cut Schottky 200 / 150 V, ≥ 3 A (PROPOSED); with it the AH 295 A-turns, the diodes 2.42 W (`sim/diodes-real-findings.md` §3.3) |
| gain | z 1.21 (z_lin, the screen's linear model); 1.139 loaded early (z_early), 1.147 with the bypass; with the 3-D utrons z_lin 1.16–1.17 (under the ≥ 1.20 rule), z_early 1.07–1.09 (`sim/utron-3d-findings.md`) |

| item | value |
|---|---|
| kick | once, at full speed (600 rpm each way, never below 500): the record's 20 % of Ψs seeds 0.110 A, 3.5 mJ; the threshold is 16 % with the deck's diodes, 17.2 % with the first-cut Schottky set, 25–28 % with silicon PN rectifiers (`sim/diodes-real-findings.md` §3.1); first cut K4, 470 µF + SCR, reed-fired (PROPOSED); with the 3-D utrons K4 starts at every phase from 1050–1100 rpm relative, the seed threshold 20–25 % (`sim/start-3d-findings.md`); with the neck's field map as well from 1150–1200 rpm relative and 25–30 % (`sim/neck-nonlinear-findings.md` §7) |
| mass | 2.23 kg per utron as built (SiFe 0.89, NiFe 0.15, Cu 1.07, G10 0.12) |
| power | belt 17.6 W (18.1 W with the bypass) + iron 1.25 W; coils 46 °C in air at a 40 °C ambient |

### AH pair (`presets/hub-locked.json`; `sim/ah-steady-cusp-findings.md`)

| item | value |
|---|---|
| cores | Fair-Rite 4077484611, 77 MnZn, Ø12.3 × 41.3 mm at ±30.7–72 mm; G-10 formers ID 13.5 / OD 16.5 mm |
| coils | 160 turns of 0.80 mm grade-1 wire in 4 layers, r 8.25–11.45 mm, ±31.35–71.35 mm; 1.01 mH; 0.333 Ω as wound (0.27 Ω in the deck); 1.22 W each, 43 °C |
| winding | two identical coils, start to the group's utrons, finish to the diode side; rotated copies, start leads toward the vessel |
| bypass (PROPOSED) | 22 mF per coil, ESR 10 mΩ [RH]; LC 34 Hz; 0.65 A rms, about 0.5 V, 8 mW for the pair |
| field | 290–308 A-turns per coil (300 mean, ±3 %; 294 as wound); top and bottom within 17 A-turns; 221–245 mean with the 3-D utrons, 249–272 with the neck's field map as well (`sim/neck-nonlinear-findings.md` §6) |
| the null | at the centre; 0.113 T/m (0.111 as wound); 25 µm per A-turn of imbalance; ±0.44 mm over a cycle with the bypass, ±6.4 mm without (`sim/ah-null-findings.md`) |
| rods | 0.121 T at 300 A-turns; 0.30 T [RH] at 743 A-turns for the AH alone |
| 77 MnZn | µi 2000; 0.51 T at 400 A/m (25 °C); Br 0.18 T; Hc 20 A/m; 1 Ω·m; Curie above 200 °C (Fair-Rite's data sheet, as a search rendered it; to verify, `presets/hub-locked.json` AH core) |

### Electrostatic pump (per side unless stated; `sim/air-stack-sizing-findings.md` §6.5, `sim/air_stack_sizing_results.json`, `sim/core_field_results.json`)

| item | value |
|---|---|
| vanes | 6 stator (REF) + 6 rotor (node 1 / 4), Al 3 mm, full rounds R1.5; 11 gaps of 6 mm air |
| sectors | 6 on a 60° pitch, 22° stator / 22° rotor; r 50–150 mm; 8° clearance each side at minimum C |
| rings of the vanes | stator ring r 150–162 mm into the G10 cage (r 162–166); rotor ring r 20.5–50 mm on the G10 sleeve (r 12.5–20.5) |
| C1 = C2 | 54.9–409.9 pF, κ 7.47 (4.81–37.3 pF per gap); as built 128.4–450.6 pF, κ 3.51: the rims add 65 / 37 pF (`sim/tube-strays-findings.md` §4) |
| Ca = Cb | 450.9 pF = 1.1 C_max: 6 full-annulus plates, 5 gaps of 6 mm, on the rotor; as laid out 480.5 pF (the gap count rounded up) |
| stack length | 102 mm of C1 + 6 mm + 48 mm of Ca = 156 mm |
| operating peak | 13.13 kV (19.7 kV breakdown / 1.5 [RH]); face field 21.9 kV/cm; rim 44.4 kV/cm (0.80 of onset) |
| gain | z 1.31 bare (1.3095 ngspice); 1.191 at start-up with the rings' chains, a large-signal gain: with real HV sticks below 1 under about 17 V (260 V hot), so a seed is needed (`sim/diodes-real-findings.md` §4.1); as built, with the tube's strays solved, 1.058 bare and 1.031 with the chains (`sim/tube-strays-findings.md` §5), and with real HV sticks a seed of 122 V / 1.0 kV / 3.65 kV (typical / maximum / hot leakage, `sim/diodes-real-findings.md` §6.3) |
| nodes | 1 / 4: −13.2 ↔ −5.7 kV (mean −8.6); 2 / 3: 0 ↔ −6.1 kV |
| clamps | Z1 / Z4 avalanche strings, BV 13.1 kV (66 × 200 V, 1.5KE200A class, PROPOSED); 1.07 W, 0.95 mA peak each |
| D1–D4 | reverse peaks 6.1 / 6.1 / 13.2 / 13.2 kV; first cut 20 kV sticks (one / two in series) + 22 kΩ (PROPOSED) |
| power | belt 2.14 W, all into the clamps; 0.65 W as built |
| aluminium (both sides) | rotor vanes 2.9 kg + Ca / Cb plates 6.1 kg; stator vanes 3.4 kg |

### Rings, supply and field (`docs/rings-design.md`; `sim/hub_rings_build_results.json`; `sim/hub_drift_results.json`; `sim/hub_revolution_results.json`)

| item | value |
|---|---|
| bands | 26.25–55.71°; 12.85 mm along the glass, 13.1 cm² each; gap 29.92 mm; Cu-ETP 0.10 mm, 12 gores per band, the laps' free edges filleted (PROPOSED) |

| item | value |
|---|---|
| beads | Ø3 mm polar (contact r 11.06, z ±22.42 mm), Ø2 mm equatorial (contact r 20.65, z ±14.08 mm) |
| ratings used | 1 kV/mm along the glass; 5 kV/mm in the gel at a bead [RH]; 2 / 8 to qualify |
| supply | 2 + 2 CW stages from the shaft; ring A −14.96 kV, ring B +14.96 kV (−14.65 / +14.67 with real HV sticks; −13.38 / +13.40 as built); 8 diodes at 7.5 kV, selected for ≤ 25 nA; 8 × 100 pF / 30 kV |
| k | 0.2546 (kV/cm)/kV as built; 0.2373 for the bands alone |
| at the null | 7.62 kV/cm, 2.57 Pa at switch-on (the bands alone 7.10); 8.20 kV/cm, 2.98 Pa settled at 25 °C; ±5 mm uniformity 4.4 % (axis) / 2.0 % (equator) |
| fields in the insulation | glass along the gap 1.00 kV/mm on average; gel at the beads 4.98 / 4.46 kV/mm as drawn; glass at the beads 1.46 / 3.01 kV/mm; to the AH 1.81 kV/mm |
| settled | gel at the beads 0.69 / 1.88 kV/mm; the PEEK 2.3 kV/mm across the seat; the equatorial beads hold while σ_glass ≤ 4.1 σ_gel |
| strays | 5.1 pF to REF each, 1.1 pF between (5.87 / 1.50 pF with the whole machine solved) |
| time | 95 % in 0.24 s (0.225 s from a consistent −1 kV seed); ripple 0.014 kV/cm p-p (0.049 with real HV sticks); one revolution 7.611–7.625 kV/cm; drift to 8.20 kV/cm in 6 h (PEEK, gel, 25 °C) |
| the hub's heat | coils 43 °C, glass 32–37 °C (32.5 °C between the rings) in a 25 °C room; the vessel 4.2e12 Ω·m there |
| leakage | 100 GΩ per ring used (ledger 158 GΩ); 4.5 mW |

**What more stages would give** (if the bench qualifies higher ratings; `sim/hub_rings_build_results.json` best_by_family)

| ratings (interface / gel) | stages a side | across | at the null (the bands alone) |
|---|---|---|---|
| **1 / 5 kV/mm (the record)** | **2** | **29.9 kV** | **7.10 kV/cm (7.62 as built)** |
| 1 / 8 | 3 | 44.3 kV | 7.80 kV/cm |
| 2 / 5 | 2 (wider bands) | 29.9 kV | 8.35 kV/cm |
| 2 / 8 | 4 | 57.9 kV | 13.38 kV/cm |

# 5. Status: decided, proposed, open

## 5.1 The record's parts by status

| part | status | note |
|---|---|---|
| the 1 : −1 gear, the belt, two pumps per side | DECIDED | the gear and belt: a first cut, PROPOSED (`docs/drive-gear-belt.md`) |
| air for the first tests; full-round vanes; the 6 + 6 cap | DECIDED | |
| 6 mm gaps, V_op = breakdown / 1.5; 22° / 22°; r 150; Ca = 1.1 C_max | DECIDED (10-09) | the designer kept the stack of record; the cost optimum differs (§3.8) |
| the clamps Z1 / Z4 | DECIDED (10-09) | |
| the HV side on the rotor; the link through one inner bearing | DECIDED | "for now"; a brush later |
| the magnetic pick (g 0.5, 6 bridges, 1200 rpm, 200 turns) | DECIDED (10-09) | the basis of DCCREG-UTR-101 (Rev A draft) |
| the steady cusp (22 mF bypass) | PROPOSED | drawn dashed on the schematic; it holds the null within ±0.44 mm |
| La / Lb, D1*–D4*, D1–D4, the clamp strings, the chains' parts, the kick source, creepage and potting | PROPOSED (first cuts) | `sim/parts-first-cut-findings.md` |
| the hub: layer order, sphere, wall, AH placement, symmetric pumps | DECIDED | |
| the rings outside the glass, copper foil, the symmetric supply | DECIDED | |
| the bands and beads | DESIGN | from the [RH] ratings; as drawn the polar bead sits 0.4 % under its rating |
| the PEEK retainer, the gel, the G10 coupler | DECIDED (materials, 10-09) | |

| part | status | note |
|---|---|---|
| the grooves' full-round tops, the AH ends rounded, the coupler as a tube, the leads' path | PROPOSED | `docs/rings-design.md` §2 |
| the joints' finish: filleted laps, seamless polar rings, dressed equatorial joints, void-free contacts | PROPOSED | `sim/hub-joints-findings.md` §5 |
| the retainer split at the equatorial plane | PROPOSED | |
| the shaft's diameter and the bearings | OPEN | the designer's choice (§5.2) |
| the ratings, the leakage, the conductivities | OPEN / ESTIMATE | the bench qualifies them |
| the multipliers stacked into the doubler (the sketch: C3–C6, D5–D8); the clamps on nodes 1 / 4; one revolving body | DECIDED (10-10) | the redesign, §3.9 |
| the redesign's circuit: bipolar, C1 / C2 in phase, the ring stages (C7 / C8, D9–D12, Csa / Csb), C3–C6 180 pF | DECIDED for now (10-11) | `sim/integrated-multiplier-findings.md`; "will do for now" |
| the redesign's layout: the decks round a grounded can, the columns toward the hub, the vane stacks against the Ca\|reluctance bearings, no hub-face bearings, the clamp strings at about 60 parts | PROPOSED | `sim/multiplier-layout-findings.md` (Figure 11) |
| the layout's bridge-ring bearings: an SKF 6205-2Z in each of the record's end slots, the end bearings one slot out (1014 mm), stainless cartridges in 8 mm G10 spiders | PROPOSED | `sim/bridge-ring-bearings-findings.md` (sheets 34–35) |

## 5.2 Open items

**For the designer to decide** (each has its numbers in this ledger):

- **the redesign** (§3.9, `sim/integrated-multiplier-findings.md`): the circuit is taken "for now" (10-11): bipolar, C1 / C2 in phase, the ring stages, C3–C6 180 pF.
  - With it: Ca / Cb at 13.2 kV; the varicaps' torque ripple tripled; no bearing link needed.
  - Its seed is 51 V / 947 V / 4.87 kV (typical / maximum / hot), against the record's 122 V / 1.0 kV / 3.65 kV below.
  - Still to run: a vane flashover with the ring stages, the torque at the gear;
- **the redesign's layout** (§3.9, Figure 11, `sim/multiplier-layout-findings.md`, PROPOSED): accept it or set it otherwise. Its consequences:
  - **the hub-face bearings go,** so the shaft needs the bridge-ring bearings, where d 40 alone gives only 26 Hz. Placed and named (an SKF 6205-2Z, 20 mm a side, 1014 mm) they give 83–87 Hz at d 25. Their preload (the record's 100 N, under SKF's 125–250 N rule) and the drive's seat at side A (§6, item 71) are the designer's;
  - **the clamp strings drop to about 60 parts,** as the layout's small strays raise the rings to ±16.25 kV at 66. A field solve of the new stack would fix the number; the bench trims it in any case;
  - **the sleeve's outer ends** need PTFE sheds (17.5 mm against 62 mm of creepage);
  - **the decks** need their parts' real envelopes and a balance;
  - **Ca / Cb's mounts** need their 13.2 kV creepage (77 mm on PTFE);
- **the shaft and its bearings** (§3.1): six bearings with d 40 (65 Hz, the runout at 0.027 mm), or an eighth pair between the shaft and each bridge ring's outer end at d 25 (131–165 Hz, the runout at 0.05 mm; the layout +10–20 mm a side; stiff seats to keep it clear of 120 Hz). The redesign's layout removes the hub-face bearings and leaves only the second way (83–89 Hz; placed and named in §3.9);
- **the drive's first cut** (§3.1, `docs/drive-gear-belt.md`): accept it, or set the gear and belt otherwise;
- **the bypass and the split**, both PROPOSED;
- **the hub's first cuts** (§3.5): the grooves' full-round tops, the AH ends rounded to 1.5 mm (which also gives the polar beads 2.3–2.4 % of margin), the coupler as a tube with the flanges turned to Ø60, the leads' path;
- **the rings' joints** (§3.6, `sim/hub-joints-findings.md` §5): the laps' free edges filleted, the polar rings seamless (turned) or their joints dressed with the AH ends rounded, the equatorial joints dressed, the contacts void-free;

- **the AH's steady field and the utrons in 3-D** (§3.2, §3.3; `sim/utron-3d-findings.md`): 300 A-turns (294 as wound) against the 449 the AH was sized to with the record's 2-D utrons, 221–245 with the 3-D ones and 249–272 with the neck's field map as well (the best estimate; z_early 1.028–1.051). The 3-D utrons' z_lin 1.16–1.17 is under the pick's own ≥ 1.20 rule. Accept the lower field; or restore it with a thicker neck (Ψs × 1.27– 1.42, about 3.8–4.3 mm of NiFe, the utrons' copper loss × 1.6–2.0), a longer stack (κ about 7.8 at 150 mm, 8.4 at 200 mm) or a larger pump; and set how each group's three coils are connected (one reversed: κ 6.44, against 5.96 aiding);
- **the neck strip** (§3.2, `sim/neck-nonlinear-findings.md`): as drawn it saturates 5 % low (a stacking factor of 0.90 [RH]) and its foils, stacked across the radius, cost 10 % of the aligned L at the laps. Keep it; stack the foils axially (the record's gain back, 322 A-turns at z_early 1.147 in the 2-D frame); or thicken it to about 3.2–3.3 mm (Ψs back). DCCREG-UTR-101 (Rev A draft) follows the choice;
- **the electrostatic pump as built** (§3.4, `sim/tube-strays-findings.md`, `sim/diodes-real-findings.md` §6): z 1.031 with the rings' chains, 0.65 W; with real HV sticks 6.82 kV/cm at the null and no self-excitation. Accepting it means the selected, cool chain parts and a seed of at least about 1 kV, or 3.65 kV for hot leakage. Or win the margin back. The cheap remedies, solved (`sim/tube-strays-findings.md` §6, [RH] options): a 6 or 12 mm radial clearance at the vanes' rims gives z 1.046 / 1.041 with the chains; a PTFE sleeve, which halves the rotor vanes' coupling to the shaft, 1.045; both, 1.059, 0.91 W and 7.15 kV/cm with ideal diodes. None comes near the record's 1.191, so the larger levers are the stack itself: more or larger vanes (the 6 + 6 cap is the designer's), insulating mounts for the Ca plates in any case;
- **the parts' first cuts** (`sim/parts-first-cut-findings.md`): La / Lb (EI-84 × 35), Schottky D1*–D4*, the HV sticks and their surge resistors, the clamp strings, the chains' parts (their diodes selected for ≤ 25 nA at 7.5 kV and kept cool), the K4 kick fired at full speed, the creepage and the potting;
- **the electrostatic pump's seed** (§3.4, `sim/diodes-real-findings.md` §4.1): with real HV sticks it does not self-excite. As built it needs 122 V on nodes 1 / 4 (typical leakage), 1.0 kV at the maximum or 3.65 kV hot, and no seed source is designed. Candidates: a one-shot HV seed fired with the magnetic pump's kick, or a tap from a charged part through an HV diode;
- **a vane flashover** (§3.4): ring B rises to 19.0 kV. Qualify the rings to ±20 kV (1.33×) on the bench, limit ring B (a spark gap near 17 kV), or rely on the vanes' 1.5 margin to keep flashovers rare;
- **the fields' purpose:** what the field at the null is for is not documented (§2.1).

**The bench** (`docs/bench-test-rings.md`):

- the hold-off ratings (phase 1), which set the stages;
- each ring's leakage (phase 2);
- the conductivities (phase 3), at 25 °C and 40 °C: the equatorial beads need σ_glass ≤ 4 σ_gel at the hub's temperature;
- the field on the pump's own supply (phase 4).

**Still open in the models and parts:**

- the 77 MnZn data sheet: its values are in `presets/hub-locked.json` (AH core), as a web search rendered the sheet; re-read them from the current sheet before ordering;
- the capacitor mounts' creepage, potting and balancing: first cuts in `sim/parts-first-cut-findings.md` and `sim/rotor-mechanics-findings.md` §6;
- the fired-on coating's edges, for the later build.

## 5.3 The bench test

Source: `docs/bench-test-rings.md`.

- **The test hub:** the sphere with a Ø8 mm pumping tube at one pole (≤ 1e-3 Pa) and aluminium AH dummies at REF.

- **The instruments:**
  - a BGO electro-optic probe on a fibre along the axis;
  - ±30 kV supplies through 100 MΩ;
  - electrometers;
  - a DC partial-discharge detector at 10 pC.
- **The phases:**
  1. **Hold-off on coupons and the sphere:** ±15.0 kV for 1 h, then ±18.7 kV for 1 h, under 10 pC. This sets the ratings and the stages.
  2. **Leakage:** at least 100 GΩ expected.
  3. **DC drift on lab supplies for 6 h:** 7.62 → 8.20 kV/cm expected, with the return after grounding and a 40 °C repeat. The ratio of the glass's conductivity to the gel's, at the hub's temperature, decides the equatorial beads once settled.
  4. **The pump's own supply at 1200 rpm relative:** as built with real HV sticks about 6.8 kV/cm, 95 % in about 0.58 s from a −1 kV seed and about 0.046 kV/cm ripple expected (`sim/diodes-real-findings.md` §6; 7.47 kV/cm on the record's capacitances, 7.62 with ideal diodes); a low reading points at the sticks' leakage. Then the AH powered, with the Faraday offset measured first.

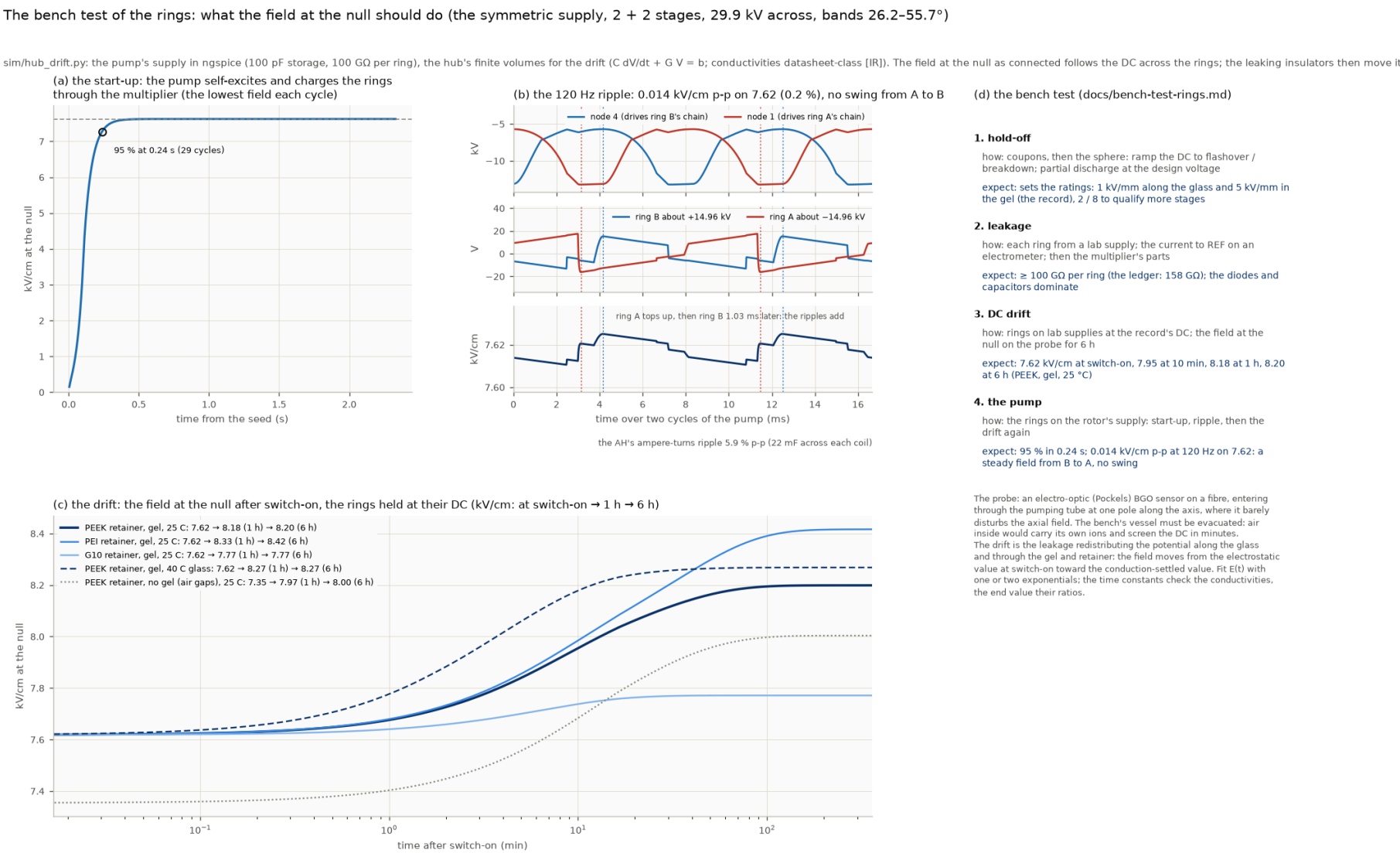



Figure 12. What the bench test should see: the start-up, the 120 Hz ripple (the pump's nodes, the rings, the field), the 6 h drift by material, and the test's phases. Source: docs/figures/hub-bench-predictions.png (docs/make_hub_bench_figure.py).

## 6. Known inconsistencies and stale records

The lock recorded 18 and fixed none. The settlement (2026-10-09) resolved them against the baseline, ran the four model checks (31–34) as studies of their own, and found 43 more (19–30 and 35–65), mostly through those studies; each line says what was done and where. Where a check corrects a number of record, the record's value stays as its basis and the corrected one stands beside it, dated, until the designer decides.

### The lock's 18

| # | what | resolution |
|---|---|---|
| 1 | the 3-D model and layout still carried the old vacuum 8 + 8 stack, Ca / Cb on the counter-rotor and the 120 mm placeholder hub | **done:** the record in solids, `sim/tube_geometry.py --record` → `docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.*` (940 mm, 153 solids, no clash); `sim/stack_sizing.py` puts the HV on the rotor |
| 2 | the shaft: d 25 in the tube, d 30 in the hub's register | **analysed, the designer's:** two bodies on the shaft (§3.1, §5.2) |
| 3 | `docs/figures/hub-locked.png` and its findings named the lock-down's rings as the record | **done:** retitled as the lock-down study; banner |
| 4 | `sim/core-field-findings.md` decisions 5 / 7, `sim/core-null-field-findings.md` called superseded designs "the record" | **done:** banners; "was the design of record" |
| 5 | the rotor schematic drew no bypass and quoted only the unbypassed A-turns | **done:** the bypass dashed (PROPOSED), both A-turn ranges, the kick box, the docstring |
| 6 | the cost sheet's BOM: 2 × 10 mF, the screen's SiFe mass, four spiders | **done:** one 22 mF part per coil, 0.89 kg, six spiders; every total +60 |
| 7 | the kick: 38 mJ reported against about 3.5 mJ seeded; two labels | **done:** `sim/pole_design.py` kick_seed reports the seeded energy, 3.48 mJ at 0.110 A, on one basis |
| 8 | A / B naming; which AH coil in which branch; the winding sense | **done:** side A below in the record (naming notes); the winding rule (`sim/ah_winding.py`) |
| 9 | the magnetic gain on two bases | **done:** named, z_lin (the screen, 1.208) and z_early (the loaded run, 1.139; 1.147 with the bypass) |
| 10 | `record_supply` E_pk_kV_cm read as the null's field | **done:** E_null_mean / pk / pp added with the convention stated |
| 11 | the stack of record against the cost optimum | **done:** the designer kept the stack of record (§2.3) |
| 12 | the η ≈ 0.45–0.50 read as the current machine's | **done:** `CONVENTIONS.md` and `README.md` place it with the disc machine |
| 13 | `README.md`, `CLAUDE.md` framed the Bennet browser tool | **done:** rewritten for the locked machine |
| 14 | `CONVENTIONS.md` switch naming: "the resonator rail" | **done:** REF, the shaft |
| 15 | `index.html`'s banner reads η ≈ 0.70 | **frozen:** flagged, not edited |
| 16 | `docs/commutator-design.md`, `docs/kicad/gap-topology-of-record.md` read as current | **done:** superseded banners |
| 17 | `CHANGELOG.md` undated; no design-loop or netlist v2–v4 entries | **done:** a dated timeline from the git history; the missing entries |
| 18 | some hub documents' tags under a redefinition | **done:** the hub documents and `presets/hub-locked.json` |

### Found and settled since

| # | what | resolution |
|---|---|---|
| 19 | the frozen list named `charge-pump-synth-live.html` and `schematic.svg` at the root; they live in `tools/` | **done:** `README.md`, `CLAUDE.md` and §8; the check prints 0 |
| 20 | **the field at the null was solved for the bands alone** | **corrected:** with the beads, 7.62 kV/cm at switch-on and 8.20 settled (§3.7); `sim/hub_rings_build.py --as-built`, `sim/hub_drift.py` and the figures carry it |
| 21 | the beads' fields came from a box filled with gel; the drawing's grooves bring the PEEK within 0.5 mm | **corrected:** 4.98 / 4.46 kV/mm as drawn (`sim/hub-beads-settled-findings.md` §3) |

| # | what | resolution |
|---|---|---|
| 22 | the AH coil: 160 turns in variant (a)'s window, its R scaled as bare copper | **settled:** 0.80 mm wire in 4 layers, 0.333 Ω; the deck keeps 0.27 Ω (`sim/hub-thermal-findings.md` §1) |
| 23 | the AH rod's "about 600 A-turns" had the cone windings in series | **corrected:** 743 for the AH alone; the code's constant stays, conservative (`sim/ah-null-findings.md` §4) |
| 24 | "all the belt's power, about 21 W" missed the windage and the bearings | **corrected:** about 39 W in all (`sim/rotor-mechanics-findings.md` §5) |
| 25 | "the coils need banding" | **corrected:** no banding; preloaded studs and bonded joints (`sim/rotor-mechanics-findings.md` §1) |
| 26 | the runout of "about 0.05 mm or better" | **corrected:** 0.027 mm total per inner bearing as laid out, 0.05 mm with the eighth pair (§3.1) |
| 27 | the air-stack study took the shaft's first mode as above 30 Hz; the earlier grounded model gave 239 Hz | **corrected:** 27–29 Hz with the counter-rotor floating (§3.1) |
| 28 | "every reaction torque goes into the gear" | **corrected:** the gear puts the same torque on both bodies; the frame takes twice it (§1) |
| 29 | the cost sheet's six bearings; the drive makes seven, the eighth pair eight | **open:** with the shaft and the drive (§5.2) |
| 30 | the hub's register says d 30, but a 6205 takes a 25 mm journal | **open:** with the shaft (§5.2) |

### **The model checks** (each a study: a script, its results and its findings in `sim/`)

| # | check | where it lands |
|---|---|---|
| 31 | the neck, a nonlinear field check | **done, and corrections:** Ψs 0.1268 Wb-turns as built (5 % low at a stacking factor of 0.90 [RH]); the aligned L 10 % lower at the laps (the foils lie across the radius); the knee sharper than the ^6 fit, so the AH gets 323 A-turns with the bypass (+8 %) for 21.0 W, in the 2-D frame (items 61–64; `sim/neck-nonlinear-findings.md`) |
| 32 | real diodes at start-up | **done, and corrections:** the magnetic pump starts on the record's 20 % kick only with Schottky rectifiers (thresholds 17.2 % for the first-cut set, 25–28 % for silicon PN); the electrostatic pump does not self-excite with real HV sticks (a seed of 17–260 V is needed, §5.2) and its rings stand at −14.65 / +14.67 kV, 7.47 kV/cm (items 46–52; `sim/diodes-real-findings.md`) |
| 33 | the tube's strays, by a field solve | **done, and a correction:** the node strays are 66.2 / 24.5 pF, not 20, and the varicap is 128–451 pF (κ 3.51), not 55–410 (κ 7.47): the vanes' rims; as built z 1.058 bare, 1.031 with the chains, 0.65 W, 7.06 kV/cm (items 53–60; `sim/tube-strays-findings.md`) |
| 34 | the utrons' κ in 3-D | **done, and a correction:** κ 6.70 in the record's frame (the ends +71 % unaligned against the [RH] +30 %), 5.96–6.44 with the coils' coupling round the machine; the pick's z_lin 1.16–1.17, under its ≥ 1.20 rule; the AH 221–245 A-turns with the bypass, not 300; the coils' connection not recorded: the designer's (`sim/utron-3d-findings.md`) |

### **Found by the parts' first cut** (`sim/parts-first-cut-findings.md`)

| # | what | resolution |
|---|---|---|
| 35 | "kick above about 200 rpm" (§3.2; `docs/drive-gear-belt.md` §4.2) | **corrected:** the pump starts at every rotor phase only from 500 rpm each way; the kick fires at full speed (`sim/parts-first-cut-findings.md` §3) |
| 36 | "seeded the other way, the pump runs mirrored" (`sim/ah-steady-cusp-findings.md`) | **corrected:** the diodes set the sign; polarised bypass parts go + to nodes d / b (§3 there) |
| 37 | D1*–D4* "silicon, 0.55 V, 2.8 A peak, reverse 112 / 101 / 43 / 45 V" | **corrected:** 0.54 V is a 100 V Schottky's drop (item 51); 1.9 / 2.3 A per diode; 113 / 104 / 58 / 60 V with the start-up (§2.1 there) |
| 38 | the Ca / Cb mounts "need 5.7–7.5 kV of creepage" | **corrected:** creepage is a length, 40 mm on PTFE or 80 mm on G10; no mount bridges two adjacent plates (§4 there) |
| 39 | La / Lb "not designed", 2.82 kg; their iron loss "negligible" | **settled:** the EI-84 × 35 first cut, 3.3 kg for the pair; 0.12–0.21 W of iron each (§1 there) |
| 40 | a kick "with a push-button" (`sim/pole-design-findings.md` §4; the cost sheet) | **corrected:** it fires at speed, so contactlessly: the K4 first cut (§3 there) |
| 41 | a vane flashover at node 4 lifts ring B to 19.0 kV, past the bench's ±18.7 kV | **open:** the designer's (§5.2) |

### **Found by the joints study** (`sim/hub-joints-findings.md`)

| # | what | resolution |
|---|---|---|
| 42 | the bead study's "a void at the contact would not discharge" took the gel-filled gap's voltage | **corrected:** it holds at switch-on only; settled, a sealed void reaching 0.5 mm from an equatorial contact breaks down (0.99 in service); the contacts are cast void-free and inspected through the glass (§4 there) |
| 43 | the gores' overlaps and joints, not modelled | **settled:** the laps hold with deburred, filleted edges and no void under them (§2 there) |
| 44 | the bead rings' closing joints: the polar beads sit 0.3–0.4 % under their rating | **settled:** the polar rings seamless, or the AH ends rounded (2.3–2.4 %); the equatorial joints dressed within δ / w 0.03 (§3 there) |
| 45 | the record's solids labelled the shaft halves and flanges "steel" against the preset's non-magnetic | **done:** "austenitic stainless, non-magnetic" (μ_r ≤ 1.05, `sim/ah-null-findings.md` §5); `sim/tube_geometry.py --record` and its GLB regenerated, the geometry unchanged |

### Found by the real-diode study (`sim/diodes-real-findings.md`)

| # | what | resolution |
|---|---|---|
| 46 | "any seed (a contact potential, a triboelectric charge) multiplies by z" (§3.4) | **corrected:** for ideal diodes only; with real HV sticks the free pump decays below about 17 V (typical leakage), 64 V (maximum) or 260 V (hot); the seed source is **open** (§5.2) |
| 47 | "95 % at 0.24 s (29 cycles)" (§3.7, §4, §5.3; `sim/hub-rings-build-findings.md` §5, `docs/rings-design.md` §6, `docs/bench-test-rings.md` phase 4) | **corrected:** the deck's `uic` start (its charge-defined varicaps start uncharged, so the −1 kV seed acts as −0.58 / −0.90 kV); 27 cycles, 0.225 s, from a consistent −1 kV (§4.2 there) |
| 48 | the rings ±14.96 kV, 57 V and 0.014 kV/cm p-p, with ideal diodes | **corrected beside the record's:** with real HV sticks −14.65 / +14.67 kV, 7.47 kV/cm and 0.049 kV/cm p-p; at the datasheet's maximum leakage 6.96 kV/cm, hot 6.20; the chain diodes selected for ≤ 25 nA and kept cool (§4.3 there) |
| 49 | `sim/diode-stack-findings.md`: the sticks' "~1 pF and μA leakage are small next to the 1 nF-class capacitors" | **corrected:** the record's chains are 100 pF; there 1 pF of body capacitance or 2 μA of leakage each cost about 2.6 kV across the gap (dated note) |
| 50 | `sim/hub_rings_build.py` LEAKAGE: a 20 kV stack "leaks tens of nA at a third of its rating" | **noted:** that is a typical or selected part; the class's datasheet maximum, 2 μA, is 80 × more (a comment there; the leakage ledger's 3e11 Ω a stage holds with selected parts) |
| 51 | "0.54 V … a Schottky-class drop"; "the threshold is 16 %" (§3.2, §4, item 37) | **corrected:** a 100 V Schottky's drop; the 200 V class that D1* / D2* need drops 0.71 V; the thresholds by class (§3.1 there) |
| 52 | `sim/pole-design-findings.md` §4: "silicon diodes (≈ 0.55 V)", "the diode drop then costs only about 2 W"; `sim/pole_design.py`'s "Si (0.55 V @ 1 A)" | **corrected:** a silicon PN rectifier drops 0.85 V at 1 A, needs a 25 % kick, and its diodes take 3.0–3.1 W; the code's label is commented, not renamed (dated notes) |

### Found by the tube-strays study (`sim/tube-strays-findings.md`)

| # | what | resolution |
|---|---|---|
| 53 | C1 = C2 "55–410 pF, κ 7.47" (§1, §3.4, §4; `sim/air_stack_sizing_results.json`): the 2-D cell with a 2 pF rim floor (`sim/stack_sizing.py` C_edge_pF) | **corrected beside the record's:** as built 128.4–450.6 pF, κ 3.51; the rims add 65 pF to C_min and 37 to C_max (dated notes; §4 there) |
| 54 | the 20 pF node stray [RH] (§3.4; `sim/core_field.py`, `sim/bicone_drive.py`) | **corrected beside the record's:** 66.2 pF on nodes 1 / 4, 24.5 pF on nodes 2 / 3 (§3 there); the decks keep their 20 pF as the record's basis (comments there) |
| 55 | Ca = Cb "451 pF = 1.1 C_max" | **corrected:** the layout rounds the gap count up, so the 5 gaps give 480.5 pF as laid out (§4 there); that is 1.07 of the solved C_max, and the rule's 1.1 would take a sixth gap, 495 pF (§6 there): the designer's |
| 56 | the 2-D cell holds the 3 mm vane as 2.4 mm node to node (`sim/stack_sizing.py`, `sim/vane_cell.py`) | **noted:** C_min per gap +7.2 % and C_max +1.0 % for the vane as drawn, the direction `sim/air-stack-sizing-findings.md` §6.5 states (comments there) |
| 57 | the rings' supply ran with the deck's 3 / 2 pF ring strays while the record quotes the hub's 5.1 / 1.1 pF | **noted:** the whole machine gives 5.87 / 1.50 pF; none of the three moves the supply (§3 there) |
| 58 | the record's solids draw C1 and C2 in phase (`sim/tube_geometry.py`: every vane at rot 0), against the circuit's antiphase | **done:** side B's stator vanes turned half a pitch (30°), as side B's bridges are; `sim/tube_geometry.py --record` re-run |
| 59 | `sim/stack_sizing.py` layout: "starting and ending on a stator vane" | **corrected (docstring):** each stack ends on a rotor vane beside the first Ca plate, the path for Ca_01's 15–23 pF |
| 60 | the parts' first cut mounts the node-1 plates on conductive spacer rings on the sleeve (`sim/parts-first-cut-findings.md` §4) | **corrected:** they would raise node 1's stray to 105 pF and take z with the chains to 1.008; insulating mounts (dated note) |

**Found by the neck check** (`sim/neck-nonlinear-findings.md`)

| # | what | resolution |
|---|---|---|
| 61 | the neck's "0.225 mWb at 0.75 T, 1.1 % above the operating point" (`sim/pole-design-findings.md` §8; DCCREG-UTR-101's data row) | **corrected (dated note):** that takes a stacking factor of 1; at 0.90 the knee is 0.211 mWb, 5 % below. The drawing's row stays until the designer settles the strip (Rev A draft) |
| 62 | "the flux crosses the strip edge-on, in the plane of the laminations" (`sim/utron_profile.py`, `sim/pole-design-findings.md`) | **corrected:** under the break only; at the laps it crosses the foils, which costs 10 % of the aligned L |
| 63 | the laps "~3 %", the knee "~5 % / ~30 %" [RH] | **corrected:** 11 %, and 12.8 % / 52 % past the knee (dated notes) |
| 64 | the pull at the clamp, 28.2 N per utron (`sim/rotor-mechanics-findings.md` §3) | **noted:** about 30 N with the field map's law in the 2-D frame (0.232 mWb aligned), about 28 N with the 3-D utrons as well, against the 834 N the studs carry at 600 rpm (`sim/rotor-mechanics-findings.md` §2) |

**Found by the as-built real-diode run** (`sim/diodes-real-findings.md` §6)

| # | what | resolution |
|---|---|---|
| 65 | as built, "the pump barely self-excites"; the seed "tens of volts, a few hundred hot", and "the deck's −1 kV is enough at every leakage level" (§3.4, §5.2) | **corrected:** those hold with ideal diodes or on the record's capacitances. As built with real HV sticks it does not self-excite: it starts from 122 V, 1.0 kV or 3.65 kV (typical, maximum, hot leakage), so −1 kV does not start it hot; 6.82 kV/cm with typical sticks. The clamps' hot leakage [RH] sets the 3.65 kV; the bench settles it |

**Found by the layout of the redesign** (`sim/multiplier-layout-findings.md`, 2026-10-11)

| # | what | resolution |
|---|---|---|
| 66 | the record's rotor wiring crosses the hub-face bearings' spiders: D3 / D4 join side A's nodes to side B's, and the rings' chains and leads join the vane bays to the hub bay. Each spider is counter-rotor, so only the shaft passes it, and nodes 1 / 4 cannot run there (3.0 pF in a bore, 0.008 of z) | **noted:** the record's layout gives no path. The redesign's layout removes the hub-face bearings (PROPOSED, §3.9); the record stays the baseline |
| 67 | the magnetic pump's leads (the AH coils in the hub to La / Lb, D1*–D4* and the utrons) cross the Ca\|reluctance spiders, unrouted (`sim/rotor-mechanics-findings.md` puts the electronics "at the hub" [RH]) | **proposed:** an 8 mm bore in each shaft half, from the flange's face under the Ca\|reluctance bearing to the reluctance bay; low voltage, 1 % of the shaft's stiffness |
| 68 | `sim/shaft_bearings.py --record` no longer ran: its density table lacked the shaft's renamed material, "austenitic stainless" | **fixed:** the key added at the record's steel figure (7850 kg/m³); the record's 28.57 Hz reproduces |
| 69 | the redesign's rings rest on the new nodes' strays, 24.5 pF each [RH] (`sim/integrated-multiplier-findings.md`) | **noted:** laid out compactly they carry 1–3 pF, and the rings rise to ±16.25 kV at the record's V_op. The layout trims the clamp strings to about 60 parts (±14.77 kV); a field solve of the new stack would settle it |

**Found by the bridge-ring bearings** (`sim/bridge-ring-bearings-findings.md`, 2026-10-11)

| # | what | resolution |
|---|---|---|
| 70 | the two-body model's rotor mass, 33.4 kg at d 25 (§4; `sim/tube-shaft-findings.md` §0) and 36.8 kg for the layout (`sim/multiplier-layout-findings.md` §7), counted the shaft halves twice: once as solids, once as the beam's steel | **corrected:** 30.4 and 33.8 kg. The model counted the shaft once, so its modes, deflections and loads stand (to 3e-8); `sim/shaft_bearings.py` fixed, `--record` and the layout study's shafts re-run, only the masses changed |
| 71 | the drive's first cut seats its gear-end bearing (the bottom bridge-ring bearing) in a hub under a 10 mm G10 end disc on bridge ring A's face, 3 mm from the utrons' end turns, on the record's 940 mm stack-up (`docs/drive-gear-belt.md` §5, §8) | **open:** the layout's seat (an 8 mm spider centred in its slot, 9 mm clear, a 6 mm ring to the face) is PROPOSED; the drive's stack-up is to be re-cut on the layout, with its frame bearing in the gear case |

## 7. Drawing register

The bundle, `docs/ledger/DCCREG-drawings-bundle.pdf`, carries every sheet below on A3, in this order. It opens with a two-page register and has bookmarks by part. The drawings stay in the repository at these paths, with the scripts that redraw them (`docs/ledger/register.py`).

**Part A — the design of record (locked)**

| sheet | title | file |
|---|---|---|
| 1 | The locked machine: what drives what | `docs/ledger/figures/architecture.png` |
| 2 | Rotor circuits: reluctance pump and electrostatic pump | `docs/schematic-rotor-circuits.svg` |
| 3 | The rings' DC supply, the symmetric pair | `docs/schematic-rings-supply.svg` |
| 4 | DCCREG-UTR-101: the wound utron's SiFe half-core | `docs/drawings/DCCREG-UTR-101.pdf` |
| 5 | The wound utron against a bridge | `docs/figures/utron-core-detail.png` |
| 6 | The air vane stack, 6 mm gaps, 6 + 6 vanes | `docs/figures/air-vane-stack-6mm.png` |
| 7 | DCCREG-HUB-201: rings A and B, copper, beaded | `docs/drawings/DCCREG-HUB-201.pdf` |
| 8 | The rings' field at the null | `docs/figures/hub-rings-field.png` |
| 9 | The machine of record: section through the shaft | `docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-section.png` |
| 10 | The machine of record: quarter cutaway | `…-hub50-3d-cutaway.png` |
| 11 | The machine of record: half section | `…-hub50-3d-half.png` |
| 12 | The locked hub in solids | `…-hub50-hub.png` |
| 13 | Reluctance sections A and B, plan cuts | `…-hub50-reluctance-plan.png` |
| 14 | Utron A1 with bridge A1, exploded | `docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-3d-exploded.png` |
| 15 | Reluctance section A, quarter cut | `…-n8-wound-g0p5-6br-3d-reluctance.png` |
| 16 | Reluctance section A, plan at mid-stack | `…-n8-wound-g0p5-6br-3d-plan.png` |

## Part B — supporting drawings and figures of the record

| sheet | title | file |
|---|---|---|
| 17 | The lock-down study of the hub | `docs/figures/hub-locked.png` |
| 18 | The rings as built: retainer, edges, stages | `docs/figures/hub-rings-build.png` |
| 19 | What the bench test should see | `docs/figures/hub-bench-predictions.png` |
| 20 | One revolution of the rotor | `docs/figures/hub-rings-revolution.png` |
| 21 | Utron size against gain: the pick | `docs/figures/utron-size-vs-gain.png` |
| 22 | The vane matrix: what the radius buys | `docs/figures/vane-matrix-radius.png` |
| 23 | The vane matrix: rim corona margins | `docs/figures/vane-matrix-corona.png` |
| 24 | The AH's null in the locked hub | `docs/figures/ah-null.png` |
| 25 | The reversing gear and the belt drive (PROPOSED) | `docs/figures/drive-gear-concept.svg` |
| 26 | The utrons in 3-D: their ends, their coupling and the pump | `docs/figures/utron-3d.png` |
| 27 | The neck in a nonlinear field solve | `docs/figures/neck-nonlinear.png` |
| 28 | Real diodes in both pumps | `docs/figures/diodes-real.png` |
| 29 | The tube's strays by a field solve | `docs/figures/tube-strays.png` |
| 30 | The multiplier stacked into the doubler (PROPOSED, the redesign of 10-10) | `docs/schematic-integrated-multiplier.svg` |
| 31 | The multiplier laid out in the machine (PROPOSED, 10-11) | `docs/figures/multiplier-layout.png` |
| 32 | The layout in solids: the section and the decks (PROPOSED, 10-11) | `docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout-section.png` |
| 33 | The layout in 3-D, its circuit at 25.5° of the cycle (PROPOSED, 10-11) | `docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50-layout-3d-bay-potential.png` |
| 34 | The bridge-ring bearings: the shafts' modes with them as drawn (SKF 6205-2Z, PROPOSED, 10-11) | `docs/figures/bridge-ring-bearings.png` |
| 35 | Side A's end in 3-D: the end bearing and the bridge-ring bearing (PROPOSED, 10-11) | `…-hub50-layout-3d-ringA.png` |

## Part C — earlier phases, superseded, kept for the record

| sheets | what | files |
|---|---|---|
| 36–38 | the machine as modelled before the record: section, cutaway, half section | `docs/geometry/tube/tube-r150-n8-wound-g0p5-6br-{section,3d-cutaway,3d-half}.png` |

| sheets | what | files |
|---|---|---|
| 39 | the first two-pump hub drive | `docs/schematic-hub-drive.svg` |
| 40–41 | the first pole pick's flux | `docs/figures/pole-pair-flux-g0p5.png`, `-g1p0.png` |
| 42–44 | the core-field studies: the pair inside, AC-coupled rings, floating cones | `docs/figures/core-null-field.png`, `core-rings.png`, `core-swing-waveforms.png` |
| 45–47 | the earlier spark-gap tube build | `docs/geometry/tube/tube-r150-n8-{section,reluctance-plan,clocking-plan}.png` |
| 48–51 | the C-EM and diode-core schematics | `docs/schematic-diode-core-{switchless,dcbus}.svg`, `docs/schematic-cem-{in-discharge-path,motor-placement}.svg` |
| 52–53 | the disc machine's KiCad schematic and its simplification | `docs/kicad/DCCREG_Turbine_circuit.svg`, `schematic_simplification.png` |
| 54–56 | the disc machine's cross-section, boomerang cap, placed motor | `tools/cross-section.svg`, `docs/boomerang-cap.png`, `docs/geometry/motor/motor-il2f-6563b90d-rc40.png` |
| 57–60 | the disc machine's field cuts | `docs/geometry/rt/slices/*.png` |

**CAD and DXF files** (not on sheets; `tools/step-viewer/` renders the STEP files):

- **of the record:**
  - `docs/drawings/DCCREG-UTR-101_half-core_A.step` and `_lamination.dxf`;
  - the record in solids: `docs/geometry/tube/tube-r150-n6-air6-wound-g0p5-6br-hub50.step`, with its `.glb` and `.parts.json`;
  - the redesign's layout in solids with its bridge-ring bearings (PROPOSED): `…-hub50-layout.step`, with its `.glb`, its `.parts.json` (each conductor's net) and the stills `…-layout-3d-{cutaway,bay,deckB}.png` and `…-3d-deckB-potential.png`; the interactive viewer with the circuit running is `tools/step-viewer/layout.html`.
- **earlier phases:**
  - the machine as modelled before the record, `tube-r150-n8-wound-g0p5-6br.step`;
  - the spark-gap tube `tube-r150-n8.step`;
  - the disc builds `freeze-v010-CaCb`, `floor-56b6cb83`, `opt-c9ac780b`, `il2-ce1a9380`, `il2f-6563b90d` (with its pump + motor);
  - the designer's C-EM source, the two disc DXF layouts (r0.15, r0.6) and the KiCad source.
- **Not drawn as manufacturing drawings:**
  - the bridge, the bridge ring, the carrier discs and cheeks, the NiFe strip and the coil as parts;
  - La / Lb (first cut in `sim/parts-first-cut-findings.md`);
  - the AH coil and core (in the record's solids);
  - the kick source (first cut in `sim/parts-first-cut-findings.md` §3);
  - the gear (its concept is sheet 25);
  - the retainer and the coupler (in the record's solids; first cut in `docs/rings-design.md` §2).

## 8. Files, tools and how to regenerate

**The records:**

- `presets/hub-locked.json` (the hub's spec);
- `sim/hub_rings_build_results.json` (record, its as-built field, record_supply, best_by_family);
- `sim/pole_design_variants_op.json` (the magnetic pick);
- `sim/air_stack_sizing_results.json` and `sim/core_field_results.json` (the electrostatic pump);
- `sim/ah_steady_cusp_results.json`, `sim/hub_drift_results.json` and `sim/hub_revolution_results.json`;
- `sim/tube_geometry_record_results.json` and `sim/shaft_bearings_record_results.json` (the record in solids, its shaft);
- `sim/integrated_multiplier_results.json` (the redesign's proposal, §3.9);
- `sim/multiplier_layout_results.json` (the redesign's layout, §3.9);

- `sim/tube_geometry_layout_results.json` and `sim/multiplier_waveforms_results.json` (the layout in solids, its circuit running, §3.9);
- `sim/bridge_ring_bearings_results.json` (the layout's bridge-ring bearings, §3.9).

**The findings,** one per study:

- `sim/pole-design-findings.md`, `sim/ah-steady-cusp-findings.md`, `sim/rotor-parts-duty-findings.md`;
- `sim/air-stack-sizing-findings.md`, `sim/vane-matrix-findings.md`, `sim/core-field-findings.md`;
- `sim/hub-locked-findings.md`, `sim/hub-rings-build-findings.md`;
- the settlement's: `sim/hub-beads-settled-findings.md`, `sim/hub-thermal-findings.md`, `sim/ah-null-findings.md`, `sim/rotor-mechanics-findings.md`, `sim/tube-shaft-findings.md` §0, `sim/parts-first-cut-findings.md`, `docs/drive-gear-belt.md`, and the model checks of §6 (31–34);
- the redesign's: `sim/integrated-multiplier-findings.md`, `sim/multiplier-layout-findings.md`, `sim/multiplier-layout-3d-findings.md`, `sim/multiplier-waveforms-findings.md` and `sim/bridge-ring-bearings-findings.md` (§3.9);
- the design and test documents `docs/rings-design.md` and `docs/bench-test-rings.md`;
- the cost guide `docs/cost/README.md`.

**Regenerate, in order:**

- **the studies:**
  - `sim/hub_rings_build.py --procs 4` (the rings; its `--as-built` step alone sets the field with the beads);
  - `sim/hub_drift.py --procs 4`;
  - `sim/hub_revolution.py`;
  - `sim/hub_beads_settled.py` and `sim/hub_thermal.py` (the hub settled, its heat);
  - `sim/ah_null.py`, `sim/rotor_mechanics.py`, `sim/drive_sizing.py --figure`, `sim/parts_first_cut.py`;
  - `sim/tube_geometry.py --record` and `sim/shaft_bearings.py --record` (the record in solids, its shaft);
  - `sim/integrated_multiplier.py --procs 4` (the redesign's proposal), then `sim/multiplier_layout.py --procs 4` (its layout), then `sim/tube_geometry.py --layout` and `sim/multiplier_waveforms.py` (the layout in solids, its circuit running), then `sim/bridge_ring_bearings.py` (its bridge-ring bearings, and `docs/figures/bridge-ring-bearings.png`; `sim/tube_geometry.py --layout` imports their placement).
- **the drawings and figures:**
  - `docs/make_rings_drawing.py`;
  - `docs/make_rings_field_figure.py`;
  - `docs/make_rings_supply_schematic.py` and `docs/make_integrated_multiplier_schematic.py`;
  - `docs/make_schematic_rotor.py`;
  - `docs/make_hub_rings_build_figure.py`, `docs/make_hub_bench_figure.py` and `docs/make_hub_revolution_figure.py`;
  - `docs/make_core_drawing.py` and `docs/make_utron_drawing.py`;
  - `docs/make_air_vane_drawing.py`;
  - `docs/make_multiplier_layout.py` (after `sim/multiplier_layout.py`);
  - the layout in 3-D: `sim/step_to_glb.py …-hub50-layout.step --angular 0.4`, then `tools/step-viewer/make_layout_data.py OUT` (the viewer's files) and `tools/step-viewer/shoot.py 8765 docs/geometry/tube/…-hub50-layout cutaway bay deckB ringA bay@0.425 deckB@0.425 --sim <OUT's sim.json>` (the stills, from a local server at the repository's root).
- **the cost:** `docs/make_cost_sheet.py`, then recalculate the workbook.
- **this ledger:** `docs/ledger/make_ledger_figures.py`, then `docs/ledger/make_ledger.py`.

**Frozen, unchanged at the lock and since:**

- `shuttle_core.py`, `reference/`, `sim/pump_engine.py`, `sim/pump_sizing.py`, `sim/pump_synth.py`, `spice/`, `index.html`, `tools/pump-calc*`, `tools/pump-synth*`, `tools/charge-pump-synth-live.html` and `tools/schematic.svg`;
- the diff against `b33baa2` is empty.

## Appendix A. Names and symbols

| name | meaning |
|---|---|
| REF, node 0 | the shaft, the rotor's electrical reference; the counter-rotor's stator vanes join it through one inner bearing |
| nodes 1 / 4 | the rotor vanes of side A / B: the electrostatic pump's pumping nodes, −13.2 ↔ −5.7 kV |
| nodes 2 / 3 | the transfer nodes between Ca / Cb and the diodes |
| C1(θ), C2(θ) | the varicaps, vanes of side A / B against REF; 55–410 pF |
| Ca, Cb | the transfer capacitors (1–2, 3–4), 451 pF |
| D1–D4 | the electrostatic doubler's diodes: D1 2→0, D2 3→0, D3 1→3, D4 4→2 |
| Z1, Z4 | the clamps on nodes 1 / 4, avalanche strings at 13.1 kV |
| A1–A3, B1–B3 | the wound utrons of side A / B; the groups L1(θ) / L2(θ) |
| La, Lb; D1*–D4*; Lp2, Lp3 | the magnetic doubler's chokes, diodes and wiring strays |
| AH | the anti-Helmholtz coil pair on the axis; the null, its centre; the cusp, its field |
| ring A, ring B | the copper bands on the vessel, A below (−15.0 kV), B above (+15.0 kV) |
| Co / Cs, Dc / Dp; Coa / Csa, Dca / Dpa | ring B's and ring A's Cockcroft-Walton parts |
| m1, m2, b1; n1, n2, a1 | the chains' oscillating and smoothing nodes |
| C3–C6; l1, l2, r1, r2 | the redesign (§3.9): the ladder's capacitors and nodes, C3 / C5 on node 2, C4 / C6 on node 3 |
| D1–D8; D9–D12, C7 / C8, xa / xb | the redesign's chains: as sketched, D1–D4 at the doubler's places and D5–D8 on the ladder; the ring stages and their nodes |
| deck A / B; the can; the column; the riser; the cross nets | the redesign's layout (§3.9): each side's potted ring of diodes and capacitors round the hub; the grounded can over the hub; Ca / Cb's plates between the deck and the vanes; node 1's (4's) lead up the column's bore; l1 and r1, the two nets that cross between the decks |
| Csa / Csb | the redesign's smoothing capacitor of ring A / B to the shaft (the record's chains number theirs Csa1, Csa2) |
| z | gain per pump cycle (the growth of the stored charge or flux) |
| κ | the swing ratio C_max / C_min or L_max / L_min |
| V_op | the electrostatic pump's operating peak, set by the clamps |
| k | the field at the null per kV across the rings, (kV/cm)/kV |
| g | a gap (never `d`, `CONVENTIONS.md` §2) |
| θ | the rotor angle (`rotor` in code; never φ, the potential) |